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## Impact des modifications de surface des implants sur les résultats à long terme du traitement chirurgical de la péri-implantite : une revue systématique

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**UNIVERSITÉ  
DE GENÈVE**



**UNIVERSITÉ  
DE GENÈVE**

**FACULTÉ DE MÉDECINE**

Clinique universitaire de médecine dentaire  
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et de premier recours  
Division de Médecine dentaire régénérative et  
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Thèse préparée sous la direction de la Professeure Catherine GIANNOPOULOU

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**Impact des modifications de surface des implants sur les résultats à long terme  
du traitement chirurgical de la péri-implantite : une revue systématique**

Thèse  
présentée à la Faculté de Médecine  
de l'Université de Genève  
pour obtenir le grade de Docteur en médecine dentaire  
par

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## RÉSUMÉ

La péri-implantite est une maladie inflammatoire affectant les tissus mous et l'os de soutien des implants dentaires et pouvant conduire, en l'absence de traitement, à la perte implantaire. De nombreuses approches chirurgicales sont utilisées pour sa prise en charge, mais l'influence des caractéristiques de surface implantaire sur le succès à long terme du traitement reste incertaine. Cette revue systématique a analysé des études cliniques humaines avec un suivi minimal de 3 ans évaluant les résultats du traitement chirurgical de la péri-implantite en fonction du type de surface implantaire. Dix-sept études ont été incluses. Globalement, les implants à surface rugueuse (modifiée) présentaient un risque plus élevé de récurrence de la péri-implantite et de perte implantaire que les implants à surface lisse (usinée/tournée). Les approches chirurgicales reconstructrices, incluant greffes osseuses et/ou membranes, semblaient offrir des résultats cliniques et radiographiques plus favorables que les approches non reconstructrices. Toutefois, la certitude des preuves était faible en raison d'une hétérogénéité méthodologique importante, de petits effectifs et d'un risque de biais. Des essais cliniques prospectifs à long terme sont nécessaires pour confirmer ces observations.

# INTRODUCTION

## Introduction

Les implants dentaires ont profondément transformé l'arsenal thérapeutique de la réhabilitation orale contemporaine en offrant une solution à la fois fiable, fonctionnelle et biologiquement compatible pour le remplacement des dents absentes. Le concept du traitement implantaire utilisant des vis en titane comme alternative aux solutions prothétiques amovibles a été introduit pour la première fois par Brånemark en 1965, marquant ainsi un véritable changement de paradigme dans le domaine de la dentisterie restauratrice. Depuis cette innovation fondatrice, la thérapeutique implanto-portée n'a cessé d'évoluer pour devenir une modalité de traitement largement acceptée et couramment utilisée, en particulier pour la réhabilitation des arcades partiellement édentées (1-4). L'adoption généralisée des implants dentaires repose sur leur capacité à restaurer la fonction masticatoire et l'esthétique de manière hautement prévisible, contribuant ainsi à une profonde transformation de la pratique clinique quotidienne.

De nombreuses études longitudinales et observationnelles ont systématiquement mis en évidence des taux de survie élevés des implants dentaires, renforçant davantage l'attrait de cette approche thérapeutique. Plusieurs investigations ont rapporté des taux de survie supérieurs à 90 % après cinq années de mise en fonction (2, 5). Plus récemment, une revue systématique regroupant des études prospectives à long terme a rapporté des taux de survie généralement supérieurs à 90 % sur une période de suivi de 5 à 10 ans, demeurant autour de 78 % après imputation à 20 ans de suivi. Par ailleurs, cinq études rétrospectives avec un suivi d'au moins 20 ans ont mis en évidence un taux de survie implantaire d'environ 88 %, même lorsque des causes de perte implantaire de nature multifactorielle étaient prises en compte (6). Pris dans leur ensemble, ces résultats confirment que les implants dentaires constituent une solution de remplacement dentaire hautement prévisible et durable.

Au-delà de ces résultats favorables en termes de survie, les réhabilitations implanto-portées présentent des avantages cliniques supplémentaires. Le traitement implantaire permet en effet de remplacer une dent absente sans nécessiter la préparation irréversible des dents adjacentes saines, comme c'est le cas pour les prothèses fixes conventionnelles (fixed dental prostheses, FDP). Cette préservation du capital dentaire, combinée à des taux de survie élevés démontrés à long terme, a largement contribué à la reconnaissance et à l'acceptation du traitement implantaire en dentisterie moderne. Néanmoins, cette approche thérapeutique n'est pas exempte d'inconvénients. Ceux-ci incluent notamment des coûts de traitement relativement élevés par rapport aux alternatives prothétiques amovibles ou fixes, ainsi que la survenue possible de complications d'ordre technique et biologique (7).

Malgré les taux de survie élevés rapportés, les complications biologiques affectant les tissus péri-implantaires sont relativement fréquentes et constituent un défi clinique majeur. Parmi celles-ci, les maladies péri-implantaires – et en particulier la péri-implantite – se sont imposées comme une menace significative pour le succès à long

terme des réhabilitations implanto-portées. La péri-implantite se caractérise par une inflammation de la muqueuse péri-implantaire associée à une perte progressive de l'os de soutien (8). Les données épidémiologiques indiquent que la péri-implantite touche environ 19,53 % des patients et 12,53 % des implants, soulignant ainsi sa prévalence considérable et son importance clinique (9).

L'étiologie de la péri-implantite est multifactorielle ; toutefois, le biofilm oral est largement reconnu comme le principal facteur étiologique initiant et maintenant le processus pathologique (10). À l'instar de la parodontite, la colonisation microbienne et la maturation du biofilm déclenchent une réponse inflammatoire de l'hôte qui conduit ultimement à la destruction tissulaire. Dans ce contexte, les interactions entre les micro-organismes et les surfaces implantaire revêtent une importance particulière, dans la mesure où elles influencent l'adhésion bactérienne précoce, le développement du biofilm et la composition microbienne. Par conséquent, les caractéristiques de surface des implants ont fait l'objet d'une attention croissante en tant que facteur potentiel impliqué dans la pathogenèse des maladies péri-implantaires.

### **Caractéristiques de surface des implants**

Les implants originaux développés par Brånemark (Nobel Biocare) étaient caractérisés par une surface dite « lisse », également qualifiée de surface usinée ou tournée. Ces implants étaient laissés inchangés après leur fabrication, à l'exception des procédures de décontamination et de stérilisation. Bien que décrites comme lisses, des observations microscopiques révèlent une surface en réalité minimalement rugueuse, résultant des traces laissées par les instruments d'usinage lors du processus de fabrication (11, 12). Pendant de nombreuses années, ces implants usinés ont constitué la référence en implantologie orale, principalement en raison de leurs résultats cliniques favorables et prévisibles (13).

Au fil du temps, des modifications de surface implantaire ont été introduites dans le but d'accélérer la cicatrisation osseuse et d'améliorer l'ancrage des implants en titane (14). Les surfaces implantaires modifiées – communément désignées comme « rugueuses » – ont été largement évaluées dans des études in vitro et in vivo (14-16). Ces modifications sont obtenues par des techniques de fabrication additives ou soustractives, qui augmentent la rugosité de surface et modifient la composition chimique superficielle (17). Dans les années 1990, des données probantes ont montré que les implants présentant un certain degré de rugosité induisaient une réponse osseuse plus importante que les implants à surface lisse (11). Ces observations ont conduit à un abandon progressif des surfaces usinées conventionnelles au profit des surfaces modifiées, qui se sont ensuite largement imposées en pratique clinique.

L'objectif principal des modifications de surface est d'améliorer la stabilité primaire et de favoriser l'ostéointégration, définie comme une connexion structurale et fonctionnelle directe entre la surface implantaire et l'os environnant (16, 18). Une ostéointégration améliorée est généralement associée à des niveaux plus élevés de contact direct os-implant (bone-to-implant contact, BIC), obtenus sur des périodes de cicatrisation plus courtes. La topographie des surfaces implantaires est classiquement divisée en macro-,

micro- et nano-topographie. La macrotopographie fait référence à la géométrie globale de l'implant, telle qu'un design conique ou cylindrique, tandis que la microtopographie correspond à la rugosité de surface à l'échelle micrométrique. Cette micro-rugosité peut être modifiée par diverses techniques, notamment l'attaque acide, le sablage ou la projection plasma (12, 16, 17, 19).

Il a été rapporté que les procédures de rugosification de surface influencent également la composition chimique des surfaces implantaires, rendant difficile la distinction entre les effets biologiques attribuables à la rugosité elle-même et ceux liés aux modifications chimiques de surface (20). Parmi les techniques additives, la projection plasma de titane (titanium plasma spraying, TPS) a historiquement été l'une des méthodes les plus utilisées. Cette technique consiste à injecter des particules de titane dans un chalumeau plasma à haute température, entraînant leur fusion sur la surface implantaire et la création d'une surface fortement rugueuse (19).

Des développements ultérieurs ont inclus la projection plasma d'hydroxyapatite (HA) et d'autres revêtements bioactifs, visant à reproduire l'environnement biochimique et l'architecture de l'os humain. Ces revêtements avaient pour objectif non seulement d'améliorer l'ostéointégration, mais également de réduire l'adhésion bactérienne et, par conséquent, le développement des maladies péri-implantaires (17, 19).

Les techniques soustractives ont également été largement adoptées. Le sablage consiste à projeter des particules céramiques sur la surface implantaire, produisant différents degrés de rugosité en fonction de la taille des particules utilisées (17). L'attaque acide, généralement réalisée à l'aide d'acides chlorhydrique ou nitrique, modifie davantage la topographie de surface. Souvent, le sablage et l'attaque acide sont combinés, donnant naissance à la surface dite SLA (sandblasted and acid-etched). Les surfaces SLA présentent des valeurs de rugosité inférieures à celles des surfaces TPS, mais supérieures à celles des surfaces usinées (11).

Plus récemment, la nano-topographie – correspondant à des structures de surface comprises entre 1 et 100 nm – a suscité un intérêt croissant au sein de la communauté scientifique. Bien que les modifications à l'échelle nanométrique soient supposées influencer le comportement cellulaire, leurs effets biologiques demeurent encore largement incompris (12, 21).

En 2004, Albrektsson et Wennerberg (11) ont proposé un système de classification des surfaces implantaires basé sur la rugosité moyenne ( $S_a$ ) : surfaces lisses ( $S_a < 0,5 \mu\text{m}$ ), minimalement rugueuses ( $S_a = 0,5\text{--}1 \mu\text{m}$ ), modérément rugueuses ( $S_a = 1\text{--}2 \mu\text{m}$ ) et rugueuses ( $S_a > 2 \mu\text{m}$ ). Bien que le terme « lisse » continue d'être employé en pratique clinique, la majorité des implants contemporains appartiennent à la catégorie des surfaces modérément rugueuses (11).

Les modifications chimiques et topographiques des surfaces implantaires ont joué un rôle central dans les performances cliniques favorables observées avec les implants modernes. Un rapport de consensus publié en 2009 a conclu que les surfaces modérément rugueuses et rugueuses présentaient une intégration osseuse plus

prononcée que les surfaces lisses et minimalement rugueuses (22). De plus, des taux de survie implantaire plus élevés ont été rapportés pour les implants à surface modifiée comparativement aux implants usinés (23, 24).

Toutefois, les données concernant le rôle des caractéristiques de surface implantaire en tant que facteur de risque des complications biologiques restent limitées. Certaines études ont suggéré que les implants présentant des surfaces très rugueuses, telles que les surfaces TPS et HA, étaient associés à une incidence plus élevée de péri-implantite (25, 26). Renvert et al. (27) ont néanmoins conclu que les données concernant l'influence des caractéristiques de surface sur l'initiation de la péri-implantite faisaient défaut, bien que des études expérimentales indiquent que les propriétés de surface puissent influencer la progression de la maladie. Des études précliniques in vivo suggèrent par ailleurs une progression plus rapide et une sévérité accrue de la péri-implantite autour des implants à surface modifiée par rapport aux surfaces tournées, ainsi que des différences entre divers types de surfaces modifiées (28, 29).

En outre, les caractéristiques de surface implantaire semblent influencer les résultats thérapeutiques après le traitement de la péri-implantite. Plusieurs études cliniques ont rapporté des résultats à court et à long terme moins favorables, ainsi que des taux de récurrence plus élevés, lors du traitement de la péri-implantite autour d'implants à surface modifiée (30-32).

## **Complications**

Les études longitudinales rapportent des taux de survie implantaire allant de 90,5 % à 100 % après un suivi d'au moins cinq ans, et de 85,5 % à 100 % après dix ans (2, 33). Malgré ces taux élevés, des échecs implantaires et des complications peuvent survenir (24). Les maladies péri-implantaires constituent l'une des complications biologiques les plus fréquemment associées au traitement implantaire (2, 34, 35).

La mucosite péri-implantaire et la péri-implantite sont regroupées sous le terme de maladies péri-implantaires. Zitzmann et Berglundh (36) ont décrit la péri-implantite comme une lésion inflammatoire destructrice d'étiologie microbienne affectant à la fois les tissus mous et durs péri-implantaires. Tandis que la mucosite péri-implantaire est limitée à la muqueuse péri-implantaire, la péri-implantite implique une perte progressive de l'os de soutien.

Bien que les maladies péri-implantaires partagent des similitudes avec la gingivite et la parodontite, des différences anatomiques et histologiques importantes existent. Les implants sont dépourvus de ligament parodontal et sont directement ankylosés à l'os. Les fibres de collagène autour des implants sont orientées verticalement et ne s'insèrent pas dans la surface implantaire, contrairement aux fibres autour des dents, qui s'insèrent horizontalement dans le ciment (37). Bien que des profils bactériens similaires soient observés dans la parodontite et la péri-implantite (10, 38), ces différences anatomiques contribuent à un tableau pathologique plus agressif dans la péri-implantite (39).

Des études histologiques ont démontré que les lésions de péri-implantite sont approximativement deux fois plus étendues que celles de la parodontite et sont en contact direct avec l'os de soutien (40). La perte osseuse progresse également plus rapidement autour des implants que des dents, en particulier autour des implants à surface rugueuse (41, 42).

### **Traitement de la péri-implantite**

L'objectif principal du traitement de la péri-implantite est d'éliminer l'inflammation, de favoriser la cicatrisation des tissus péri-implantaires et de restaurer les structures et fonctions perdues (43, 44). De nombreuses revues systématiques ont abordé le traitement de la péri-implantite (45-50), mettant en évidence une hétérogénéité considérable des protocoles thérapeutiques et des résultats.

La thérapie non chirurgicale seule s'est révélée parfois insuffisante pour le traitement de la péri-implantite (44, 49), ce qui a conduit à l'approche chirurgicale (44, 46). Les interventions chirurgicales offrent un accès direct aux défauts péri-implantaires, permettant la décontamination de la surface implantaire et, lorsque cela est indiqué, la mise en œuvre de procédures régénératives. Des stratégies chirurgicales non régénératives et régénératives ont été décrites, incluant notamment l'implantoplastie, l'ostéoplastie, la chirurgie résectrice et la régénération osseuse guidée (guided bone regeneration, GBR).

Malgré l'abondance des recherches, aucune méthode de décontamination de surface n'a démontré une supériorité claire (47), et la prévisibilité à long terme des approches régénératives demeure incertaine (51). Il est particulièrement notable que, bien que de nombreuses revues aient évalué les différentes modalités chirurgicales, aucune ne se soit spécifiquement concentrée sur l'influence de la topographie de surface implantaire sur les résultats du traitement chirurgical de la péri-implantite.

Malgré plusieurs décennies de recherche consacrées aux modifications de surface implantaire, à la pathogenèse des maladies péri-implantaires et aux stratégies chirurgicales de traitement, une lacune majeure persiste concernant le rôle de la topographie de surface implantaire dans la détermination des résultats thérapeutiques après un traitement chirurgical de la péri-implantite. La littérature existante s'est principalement concentrée sur la survie implantaire, l'ostéointégration ou l'efficacité globale des traitements, sans intégrer de manière systématique les caractéristiques de surface comme facteur modulateur des résultats cliniques et radiographiques. En outre, les données disponibles demeurent fragmentées et hétérogènes, limitant ainsi la possibilité de tirer des conclusions définitives et de guider la prise de décision clinique.

L'originalité du présent travail doctoral réside dans son évaluation ciblée et systématique de la topographie de surface implantaire en tant que déterminant des résultats du traitement chirurgical de la péri-implantite chez l'être humain. En synthétisant les données cliniques et radiographiques issues de différentes catégories de surfaces implantaires, ce travail vise à déterminer si les implants à surface rugueuse et lisse présentent des schémas de cicatrisation, des réponses thérapeutiques et une stabilité à

long terme distincts après une intervention chirurgicale. Ce faisant, cette thèse ambitionne de combler un fossé critique entre l'ingénierie des surfaces implantaires et la gestion clinique des maladies péri-implantaires, en apportant des données originales et cliniquement pertinentes susceptibles d'orienter le choix des implants, la planification thérapeutique et les stratégies de maintenance à long terme chez les patients atteints de péri-implantite.

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# Impact of implant surface modifications on long-term outcome of surgical peri-implantitis treatment: a systematic review

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**Introduction:** Peri-implantitis is an inflammatory disease that compromises peri-implant tissues and supporting bone, potentially leading to implant loss. Although several surgical treatment strategies have been proposed, it remains unclear whether implant surface characteristics (smooth vs. rough) influence long-term treatment outcomes.

**Methods:** A systematic review was conducted to evaluate clinical studies with a minimum follow-up of 3 years that assessed the outcomes of surgical treatment of peri-implantitis in relation to implant surface type. Data extraction focused on recurrence of peri-implantitis, implant survival, clinical parameters, radiographic outcomes, and the type of surgical approach used (reconstructive vs. non-reconstructive).

**Results:** Seventeen clinical studies were included. Outcomes varied according to implant surface characteristics. Rough (modified) surfaces were generally associated with a higher risk of recurrence of peri-implantitis and implant loss compared with smooth (machined/turned) surfaces. Reconstructive surgical approaches, especially those involving bone grafts and membranes, demonstrated more favorable outcomes compared with non-reconstructive approaches.

**Discussion:** Despite observed trends, the certainty of the evidence remains low due to heterogeneity between studies, small sample sizes, and methodological limitations. Further well-designed long-term clinical trials are needed to clarify the role of implant surface characteristics in the long-term success of peri-implantitis surgical treatment.

**Systematic Review Registration:** PROSPERO (CRD420251129791).

## KEYWORDS

peri-implantitis, surgical peri-implantitis treatment, treatment outcome, explantation, implant surface, long-term outcomes, bone loss, implant survival

# Impact of Implant Surface Modifications on Long-Term Outcome of Surgical Peri-Implantitis Treatment: A Systematic Review

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**Keywords:** Peri-implantitis, surgical peri-implant treatment, treatment outcome, explantation, implant surface, long-term outcomes, bone loss, implant survival, disease recurrence

**Abstract**

Peri-implantitis is an inflammatory disease that compromises peri-implant tissues and supporting bone, potentially leading to implant loss. Although several surgical treatment strategies have been proposed, it remains unclear whether implant surface characteristics (smooth vs. rough) influence long-term treatment outcomes.

**Methods:** A systematic review was conducted to evaluate clinical studies with a minimum follow-up of 3 years that assessed the outcomes of surgical treatment of peri-implantitis in relation to implant surface type. Data extraction focused on recurrence of peri-implantitis, implant survival, clinical parameters, radiographic outcomes, and the type of surgical approach used (reconstructive vs. non-reconstructive).

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**Discussion:** Despite observed trends, the certainty of the evidence remains low due to heterogeneity between studies, small sample sizes, and methodological limitations. Further well-designed long-term clinical trials are needed to clarify the role of implant surface characteristics in the long-term success of peri-implantitis surgical treatment.

## 1. Introduction

Dental implants have significantly advanced oral rehabilitation, providing highly predictable solutions for tooth replacement. For instance, a recent systematic review reported that long-term prospective studies on dental implants show high survival rates, typically exceeding 90% over 5–10 years and remaining around 78% after imputation at 20 years follow-up. In addition, five retrospective studies with  $\geq 20$  years of follow-up reported an implant survival rate of approximately 88%, including multifactorial causes(6).

However, despite the high survival rates, biological complications at implants are rather common. In particular, peri-implantitis, which is characterized by peri-implant mucosal inflammation and progressive bone loss, affects approximately 19.53% of patients and 12.53% of implants, highlighting its relevance in clinical practice (9). As the main etiological factor for peri-implantitis is the oral biofilm, microbial to implant surface interactions seem to play an important role in disease pathogenesis. Indeed, surface modifications (e.g., sandblasting, acid-etching, anodization, etc) aiming in enhancing implant surface bioactivity, substantially impact on microbial colonization and biofilm development (28, 29, 31, 52, 53). Indeed, although the incidence of peri-implantitis seems not to differ between modified and non-modified (i.e., turned) implants in the clinic, progression and severity of peri-implantitis appear linked to implant surface properties; specifically, pre-clinical *in vivo* studies indicate a faster disease progression at modified implants compared with turned implants, as well as differences in disease progression among various modified surfaces (28, 29). Moreover it seems that implant surface characteristics may impact on treatment outcomes both in the short-term but also on the long-term, with implants with a modified surface demonstrating less positive results and higher recurrence rates (30-32).

Despite technological advancements and improved treatment approaches, the impact of implant surface modifications on peri-implantitis outcomes remains unclear. Therefore, this systematic review aims to evaluate whether varying implant surface topographies influence clinical and radiographic outcomes following surgical peri-implantitis treatment in humans. The findings may offer critical insights guiding the selection of implant surface characteristics to enhance treatment efficacy.

## 2. Materials and methods

### Study Design

This review was performed following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and was registered in PROSPERO (ID: CRD420251129791.).

### Search Strategy

To identify relevant studies, we systematically searched PubMed, Embase, and the Cochrane Library. The search strategy was carried out in English language from database inception for articles published between 2000 to 2025. Two investigators (AZ and PG) independently reviewed the search results and screened the titles and abstracts. Full texts of all potentially eligible studies were obtained. In PubMed, the following search strategy was used: "(Periimplantitis OR peri-implantitis OR peri implantitis OR periimplant OR peri-implant OR peri implant) AND (treatment outcome OR therapy OR surgical treatment OR regenerative OR regeneration OR tissue

regeneration OR reconstructive surgery OR bone graft OR bone substitute OR membranes OR surgical flap OR open flap debridement OR resective OR implantoplasty OR surface decontamination) AND (surface characteristics OR surface roughness OR material characteristics OR titanium surface OR implant types OR implant surfaces OR surface topography OR surface analysis) AND (implant survival OR bone loss OR recurrence OR retreatment OR radiographic stability OR long-term OR 3 years OR follow-up)." This search strategy was adapted to suit the other electronic sources. The reference lists of retrieved articles were also checked to identify additional studies of interest. Any inconsistencies were resolved by consensus with a third investigator (CG). The complete search strategies for all databases are provided in Appendix 2.

### **Criteria for Considering Studies for This Review**

*Study design:* Randomized controlled trials, prospective studies, retrospective studies, case-control studies, and case series were included. No specific cut-off criteria for sample size were applied, given the limited availability of data. Additionally, two case series with very small sample sizes were included due to their clinical relevance. Eligibility required that included studies explicitly reported the implant surface type(s) of the implants investigated.

*Population:* Human studies. Patients with osseointegrated dental implants diagnosed with peri-implantitis, treated surgically, with a follow-up period of at least 3 years (or an average  $\geq 3$  years).

*Intervention:* Surgical therapy for peri-implantitis.

*Comparator:* Different implant surface types, characterized by variations in macro-, micro-, and nano-scale surface roughness, topography, and material composition. Surfaces were categorized as non-modified (i.e. turned, smooth, machined), modified (rough), or mixed (hybrid), depending on their reported surface characteristics.

#### *Outcomes:*

##### Primary Outcome:

- Percentage of implants with recurrence of peri-implantitis requiring re-treatment or explantation or simply defined as treatment failure by the authors.

##### Secondary Outcomes:

- Implant loss (due to any reason)
- Disease resolution defined by reduction of probing depth (PD) without bleeding on probing (BOP) or suppuration
- Radiographic bone loss or gain assessed by mean changes in bone levels or percentage of implants with stable bone levels post-treatment
- Mean probing depth (PD) post-treatment

Subgroup synthesis: The outcomes were further stratified based on implant surface types and surgical approach:

1. Turned (machined/non-modified surfaces)
  - a. Non-reconstructive surgical approach
  - b. Reconstructive surgical approach (regardless the technique or materials used)
2. Modified (rough surfaces)
  - a. Non-reconstructive surgical approach

- b. Reconstructive surgical approach (regardless the technique or materials used )
- 3. Mixed or unspecified surfaces
  - a. Non-reconstructive surgical approach
  - b. Reconstructive surgical approach (regardless the technique or materials used )

**Data Collection** Two investigators independently extracted key data from the included articles. The inter-rater agreement for study selection was assessed using Cohen's kappa statistics. Inter-rater reliability was assessed using Cohen's kappa statistic on a subset of 20% of studies, yielding a kappa of 0.85, indicating a high level of agreement. Discrepancies were resolved through discussion or consultation with a third reviewer (CG). For each article, we extracted study features (i.e., study design, year of publication, number of enrolled patients), type of intervention, and outcome measures were extracted. Correct data extraction was controlled in a subset of randomly selected studies by the third investigator.

### **Assessment of Risk of Bias**

Two investigators independently appraised the risk of bias of the included studies using the Cochrane Risk of Bias Tool 2.0 (RoB2) for RCTs. For non-RCTs the Risk of Bias in Non-randomized Studies of Interventions (ROBINS-I) tool was used. Any inconsistencies were resolved by consensus with a third investigator (CG).

### **Data Synthesis**

Preliminary analyses of available data revealed high heterogeneity, precluding meaningful meta-analysis. Therefore, a narrative synthesis was conducted. These limitations included significant heterogeneity in implant surface types, surgical techniques, and reported outcome measures across studies. To enhance clarity and readability, findings were systematically summarized in tables according to pre-defined outcomes and subgroup analyses.

Data extraction was performed separately for each treatment group within studies containing multiple groups, while data from studies with a single treatment group were extracted accordingly. Results were categorized based on implant surface types and surgical approaches as follows:

1. Turned (machined/non-modified surfaces)
  - a. Non-reconstructive surgical approach
  - b. Reconstructive surgical approach (regardless the technique or materials used)
2. Modified (rough surfaces)
  - a. Non-reconstructive surgical approach
  - b. Reconstructive surgical approach (regardless the technique or materials used)
3. Mixed or unspecified surfaces
  - a. Non-reconstructive surgical approach
  - b. Reconstructive surgical approach (regardless the technique or materials used)

Findings were systematically summarized in tables according to pre-defined outcomes and subgroup analyses to enhance clarity and readability.

Within each treatment group, data were systematically collected on key parameters, including sample size (number of participants and implants), criteria used to define peri-

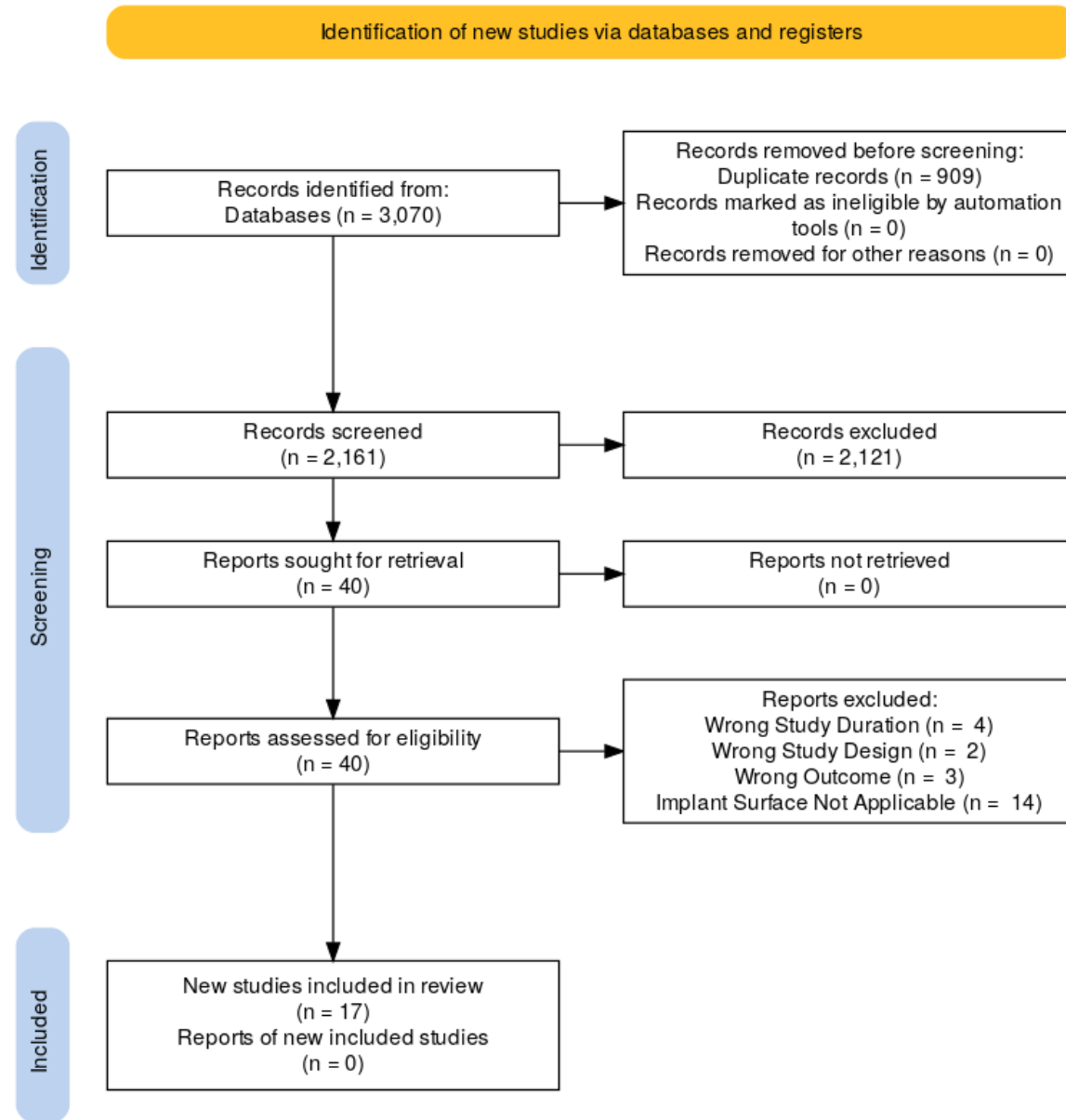
implantitis, type of bone substitute, membrane used (if applicable), follow-up periods, implant system, and implant surface characteristics.

### **3. Results**

#### **Study selection**

The literature search process is illustrated in the flowchart below (Figure 1). In total, there are 17 studies included in the analysis (30, 32, 54-68). Among them, 8 are prospective cohort studies, 3 are retrospective cohort studies, 1 are randomized controlled trials. The remaining studies include 1 each of the following types: prospective clinical study, retrospective observational study, and prospective case series. A detailed description of the study characteristics can be found in the results in Tables 2a and 2b.

**Figure 1. PRISMA Flow Diagram**



## **Study populations**

### **Peri-implantitis**

Across the 17 studies analyzed, various diagnostic criteria have been employed to identify peri-implantitis, reflecting differences in study designs and clinical considerations (30, 32, 54-68). The most commonly reported diagnostic parameters include probing depth (PD), bleeding on probing (BOP), suppuration, and radiographic evidence of bone loss.

#### *Probing Depth (PD):*

A probing depth threshold of  $\geq 6$  mm is frequently used as a criterion to identify peri-implantitis, as observed in studies by Carcuac et al. (2020), Romandini et al. (2023), and Roccuzzo et al. (2017, 2020, 2021) (15, 16, 26-28). Other studies, such as Aghazadeh et al. (2022) and Noelken et al. (2023), set a threshold of  $\geq 5$  mm, which is similar to the  $> 4$  mm threshold considered indicative of disease by Mercado et al. (2018) and Schwarz et al. (2009) (24, 25, 31, 32). This variation highlights differences at diagnosis across studies.

#### *Bleeding on Probing (BOP) and Suppuration:*

The presence of BOP and/or suppuration was consistently reported as a diagnostic marker in all studies. It serves as an indicator of ongoing inflammation and peri-implant tissue destruction. Studies such as La Monaca et al. (2018, 2024) and Khoury & Buchmann (2001) emphasize the importance of these clinical signs in combination with radiographic findings for accurate diagnosis (20-22).

#### *Radiographic Bone Loss:*

Radiographic evaluation of bone loss is another widely accepted criterion for peri-implantitis diagnosis. The threshold for bone loss varies among studies, with the most commonly reported value being  $\geq 3$  mm, as seen in Romandini et al. (2023), and (16). Other studies, including Carcuac et al. (2020) and Aghazadeh et al. (2022), defined progressive bone loss based on post-treatment changes or specific defect characteristics, such as angular defects of  $\geq 3$  mm (15, 32). A more conservative threshold of  $\geq 1.8$  mm was applied in studies such as Roos-Jansåker et al. (2011, 2014), reflecting the variability in bone loss progression (29, 30).

#### *Variability in Diagnostic Criteria:*

Despite a general agreement on the primary diagnostic signs—probing depth, BOP/suppuration, and radiographic bone loss—variability exists in the specific thresholds and additional criteria applied across studies.

A detailed overview of the case definitions used to include patients with peri-implantitis in each study (treatment group) is provided in Table 1.

**Table 1: Definitions of periimplantitis**

| Study                                  | Definition of periimplantitis                                                                                                                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aghazadeh et al., 2022 (32)</b>     | Probing pocket depths of at least 5 mm, presence of bleeding on probing and/or suppuration, radiographic bone loss of 2 mm or more from implant placement to screening, and an angular peri-implant bone defect of 3 mm or greater |
| <b>Carcuac et al., 2020 (15)</b>       | Probing pocket depths of 6 mm or more, presence of bleeding on probing, reduced marginal bone level and progressive bone loss greater than 1 mm post-treatment                                                                     |
| <b>Deppe et al., 2007 (18)</b>         | Probing pocket depths of at least 5 mm, presence of bleeding on probing, radiographic evidence of progressive vertical bone loss, and clinical signs of inflammation                                                               |
| <b>Jemt et al., 2021 (19)</b>          | Bone loss exceeding 0.4 mm, mucosal inflammation, presence of plaque and/or suppuration, and radiographic evidence of marginal bone loss                                                                                           |
| <b>Khoury et al., 2001 (20)</b>        | Bone loss of more than 50% of the implant length, augmented probing depths, bleeding on probing, and radiographic evidence of intrabony defects                                                                                    |
| <b>La Monaca et al., 2018 (21)</b>     | Progressive bone loss of 3 mm or more detected on radiographs, the presence of bleeding on probing and/or suppuration, and probing depths of at least 5 mm                                                                         |
| <b>La Monaca et al., 2024 (22)</b>     | Progressive angular bone loss of at least 3 mm beyond crestal bone level changes, the presence of bleeding on gentle probing and/or suppuration, and implants in function for more than 12 months                                  |
| <b>Leonhardt et al., 2003 (23)</b>     | Marginal bone loss of at least three implant threads compared to baseline radiographs, bleeding on probing and/or suppuration from peri-implant sulci, and microbiological confirmation of peri-implant pathogens                  |
| <b>Mercado et al., 2018 (24)</b>       | Probing pocket depths exceeding 4 mm, the presence of bleeding on probing and/or suppuration, a minimum radiographic bone loss of 20%, and implants that have been in function for at least 2 years                                |
| <b>Noelken et al., 2023 (25)</b>       | Probing pocket depths greater than 5 mm, the presence of bleeding on probing and suppuration, and radiographically confirmed bone loss                                                                                             |
| <b>Roccuzzo et al., 2017 (26)</b>      | Probing pocket depths of at least 6 mm, no implant mobility, bleeding on probing and/or suppuration, and radiographic bone loss exceeding three implant threads compared to baseline                                               |
| <b>Roccuzzo et al., 2020 (27)</b>      | Probing pocket depths of 6 mm or greater, the presence of bleeding on probing and/or suppuration, radiographic bone loss beyond crestal changes, and the absence of implant mobility                                               |
| <b>Roccuzzo et al., 2021 (28)</b>      | Probing pocket depths reach or exceed 6 mm, bleeding on probing, radiographic evidence of progressive bone loss, and the presence of pus or inflammation                                                                           |
| <b>Romandini et al., 2024 (16)</b>     | Probing pocket depths of 6 mm or more, bleeding and/or suppuration on probing, and radiographic evidence of marginal bone loss equal to or greater than 3 mm                                                                       |
| <b>Roos-Jansåker et al., 2011 (29)</b> | Radiographic bone loss of at least 1.8 mm following the first year in function, the presence of bleeding and/or pus on probing, and inclusion criteria of non-mobile implants                                                      |
| <b>Roos-Jansåker et al., 2014 (30)</b> | Radiographic bone loss of at least 3 threads ( $\geq 1.8$ mm), the presence of a vertical defect component, and bleeding on probing and/or suppuration                                                                             |
| <b>Schwarz et al., 2009 (31)</b>       | Probing pocket depths greater than 4 mm, presence of bleeding on probing and/or suppuration, radiographic evidence of bone loss, and an intrabony defect component of at least 3 mm                                                |

### **Primary Outcome: Recurrence and Treatment Failure**

The included studies demonstrated that implant surface characteristics influenced recurrence rates following surgical peri-implantitis treatment. Modified (rough) surfaces consistently showed higher recurrence compared with turned (machined) surfaces. Carcuac et al. (2020) reported an overall recurrence of 44%, with a significantly increased risk for modified surfaces (OR 5.1) (15). Similarly, Romandini et al. (2024) found a retreatment rate of 24.3%. In contrast, studies involving turned surfaces, such as Leonhardt et al. (2003), reported more stable outcomes (23). These findings indicate that surface roughness is a key determinant of recurrence and long-term treatment stability. Studies by Schwarz et al. (2009), Mercado et al. (2018), and Noelken et al. (2023) documented relatively stable outcomes without explicitly reporting significant recurrence rates (24, 25, 31).

### **Secondary Outcomes**

#### **Implant Loss:**

Implant loss was more frequent among rough surface implants, especially TPS, with Roccuzzo et al. (2020) reporting loss in 45% of TPS implants (27), and Leonhardt et al. (2003) reporting 27% for turned surfaces (23). SLA surfaces demonstrated better survival than TPS, with 20% vs. 45% loss after 10 years (27). Modified surfaces were identified as a strong predictor of implant loss (HR 4.5)(16). Turned surfaces generally exhibited lower long-term loss risk, around 20%, compared to modified ones (16).. Lower implant loss rates were generally associated with reconstructive surgical approaches, as observed by Noelken et al. (2023, 8.3%) (25) and La Monaca et al. (2024) (22), 8.8%.

#### **Disease Resolution and Probing Depth (PD):**

Reconstructive surgery generally improved PD irrespective of surface, but rough surfaces demonstrated greater variability. Mercado et al. (2018) reported PD reduction from 8.9 mm to 3.5 mm on micro-rough implants (24) , while Noelken et al. (2023) achieved PD reduction from 5.05 mm to 3.08 mm in predominantly rough implants(25) identifying disease resolution. Roccuzzo et al. (2017, 2020) observed significant PD improvements for SLA implants compared with TPS (26, 27). Turned implants (Leonhardt et al., 2003) also demonstrated significant PD reduction (23). Conversely, non-reconstructive surgical approaches such as that of Deppe et al. (2007) with predominantly rough surfaces showed initial short-term PD reductions with inconsistent long-term stability (18).

#### **Radiographic Bone Changes:**

Bone regeneration outcomes were surface-dependent. Reconstructive procedures around rough implants, particularly SLA, showed consistent bone gain (Roccuzzo et al. 2017: +2.1 mm; 2020: +2.7 mm) (26, 27). TPS implants demonstrated less favorable long-term stability, even with grafting (27). Smooth (turned) surfaces were rarely evaluated in regenerative contexts, limiting conclusions. Khoury & Buchmann (2001) reported substantial bone gain (3.2 mm) on rough implants with autografts (20) , while Roos-Jansåker et al. (2011, 2014) found stable bone gain (1.1–1.6 mm) in predominantly machined implants (29, 30).

### **Subgroup Analyses**

**Turned surfaces:** Showed moderate long-term stability, but implant loss remained high when treated non-reconstructively (23). Reconstructive data were limited but suggested stable outcomes (29, 30).

**Modified (rough) surfaces:** Non-reconstructive approaches resulted in high recurrence and implant loss (15, 16). Reconstructive approaches improved outcomes, with SLA surfaces outperforming TPS (26, 27).

**Mixed surfaces:** Outcomes were heterogeneous. Laser-assisted non-reconstructive therapy demonstrated short-term benefits (Deppe et al. 2007) (18), but Jemt & Eriksson (2021) reported long-term bone loss regardless of surface type (19). Reconstructive treatments showed better results with natural bone mineral combined with a collagen membrane (NBM+CM) compared to nanocrystalline hydroxyapatite (NHA) (31), but surface-specific differences remained underreported. Aghazadeh et al. (2022) reported improved outcomes with xenograft (BDX) usage (32).

The detailed study characteristics and outcomes are presented in Tables 2a and 2b.

**Table 2a: Summary of clinical studies evaluating treatments of peri-implantitis: surface types, materials, and outcomes**

| Author (Year)                            | Implant Surface                              | Implant Brand                                                    | Patients (initial/follow-up) | Implants (initial/follow-up)   | Type of Treatment                                                                       | Materials Used                                                                                                                            | Follow-up Period | Surface Impact                                                             |
|------------------------------------------|----------------------------------------------|------------------------------------------------------------------|------------------------------|--------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------|
| <b>Aghazadeh et al. (2022) (32)</b>      | Mixed (Turned and medium-rough)              | Not explicitly stated, categorized by surface type               | 45 (AB:22/16, BDx:23/23)     | 75 (AB:36/25, BDx:39/38)       | Reconstructive surgery (autogenous bone, xenograft, collagen membrane)                  | Autogenous bone, xenograft (Bio-Oss), collagen membrane (OsseoGuard), titanium curettes, hydrogen peroxide, azithromycin, CHX rinse       | 5 years          | No significant difference (turned vs medium-rough surfaces)                |
| <b>Carcuac et al. (2020) (15)</b>        | Mixed (modified/non-modified)                | Not specified, classified as non-modified/modified               | 100/73                       | 179/130                        | Open-flap debridement, surface decontamination, pocket elimination                      | Systemic antibiotics, local antiseptics                                                                                                   | 5 years          | Modified surfaces higher recurrence risk (OR 5.1, 95% CI: 1.6–16.5)        |
| <b>Deppe et al. (2007) (18)</b>          | Mixed (rough predominantly)                  | IMZ, Frialit-2, Brånemark, Straumann screw-type                  | 32                           | 73 (Conventional:34, Laser:39) | Conventional vs CO <sub>2</sub> laser-assisted (soft tissue resection/augmentation)     | CO <sub>2</sub> laser, air-powder abrasive, $\beta$ -TCP/autogenous bone, Gore-Tex membrane                                               | 5 years          | Superior outcomes with CO <sub>2</sub> laser in non-reconstructive therapy |
| <b>Jemt &amp; Eriksson (2021) (19)</b>   | Mixed (Turned and moderately rough surfaces) | Brånemark (turned), TiUnite, Astra Tech OsseoSpeed, Lifecore RBM | 122 (initially 134, 12 lost) | 614/453                        | Non-reconstructive surgery (mechanical cleaning, osseous recontouring, antibiotics)     | Mechanical debridement, hydrogen peroxide 10%, systemic antibiotics                                                                       | Mean 7.3 years   | No significant difference (turned vs moderately rough surfaces)            |
| <b>Khoury &amp; Buchmann (2001) (20)</b> | Rough surfaces                               | IMZ, Frialit-2 (Friadent GmbH)                                   | 25                           | 41                             | Reconstructive surgery (autogenous bone $\pm$ membranes)                                | Autogenous bone, ePTFE/Bioabsorbable barriers, CHX, citric acid, hydrogen peroxide, systemic antibiotics                                  | 3 years          | No surface comparison (all rough)                                          |
| <b>La Monaca et al. (2024) (22)</b>      | Rough (TiUnite surface)                      | Nobel Biocare (Brånemark System, Göteborg)                       | 34/23                        | 34/20                          | Reconstructive surgery (MDBA, resorbable membrane, chemical/mechanical decontamination) | MDBA (Puros), collagen membrane (Bio-Gide), hydrogen peroxide, CHX solution, tetracycline hydrochloride, systemic antibiotics (Amox/clav, | 10 years         | All rough (TiUnite); no surface-specific comparison                        |

|                                     |                             |                                                                                              |       |       |                                                                                                           |                                                                                                                                     |           |                                                     |
|-------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------|-------|-------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------|
|                                     |                             |                                                                                              |       |       |                                                                                                           | Metronidazole)                                                                                                                      |           |                                                     |
| <b>La Monaca et al. (2018) (21)</b> | Rough (TiUnite surface)     | Nobel Biocare (Brånemark System, Göteborg)                                                   | 34    | 34    | Reconstructive surgical therapy (MDBA, resorbable collagen membrane, chemical/mechanical decontamination) | MDBA (Puros), resorbable membrane (Bio-Gide), hydrogen peroxide, CHX 0.2%, tetracycline hydrochloride, systemic antibiotics         | 5 years   | All rough (TiUnite); no surface comparison          |
| <b>Leonhardt et al. (2003) (23)</b> | Turned surfaces             | Brånemark System (Nobel Biocare)                                                             | 9     | 26    | Surgical non-reconstructive therapy + systemic antibiotics                                                | Hydrogen peroxide 10%, individualized antibiotics (metronidazole, amoxicillin, tetracycline, ciprofloxacin, clindamycin), CHX rinse | 5 years   | All turned surfaces; not directly analyzed          |
| <b>Mercado et al. (2018) (24)</b>   | Rough (Micro-rough)         | Branemark TiUnite (46.66%), Astra Tech (26.66%), Straumann (10%), Others (16.66%)            | 30    | 30    | Regenerative surgery (DBBMC, EMD, doxycycline, EDTA, ultrasonic scaler)                                   | DBBMC, EMD, doxycycline, EDTA 24%, ultrasonic scaler, chlorhexidine 0.12%                                                           | 36 months | Not specifically analyzed (micro-rough surfaces)    |
| <b>Noelken et al. (2023) (25)</b>   | Mixed (rough predominantly) | Straumann, Ankylos, Brånemark, NobelActive, NobelPerfect, Frialit I, OsseoSpeed, Camlog, ICX | 18    | 24    | LAPIDER (laser-assisted regeneration, autogenous bone, CT graft)                                          | Er:YAG laser, autogenous bone chips, doxycycline, CT graft, resorbable sutures                                                      | 3 years   | No direct comparison; rough surfaces only           |
| <b>Roccuzzo et al. (2017) (26)</b>  | Rough (SLA and TPS)         | Straumann (SLA, TPS)                                                                         | 26/24 | 26/24 | Regenerative surgery (DBBMC, EDTA, CHX)                                                                   | DBBMC (Bio-Oss Collagen), EDTA 24%, CHX gel 1%, antibiotics (Amoxicillin/clavulanic acid)                                           | 7 years   | SLA better clinical outcomes than TPS (significant) |
| <b>Roccuzzo et al. (2020) (27)</b>  | Rough (SLA and TPS)         | Straumann (SLA, TPS)                                                                         | 26/14 | 26/14 | Regenerative surgery (DBBMC, EDTA, CHX)                                                                   | DBBMC (Bio-Oss Collagen), EDTA 24%, CHX gel 1%, antibiotics (Amoxicillin/clavulanic acid)                                           | 10 years  | SLA superior survival/outcomes compared to TPS      |
| <b>Roccuzzo et al. (2021) (28)</b>  | Rough (SLA)                 | Straumann (SLA implants)                                                                     | 75/51 | 75/64 | Reconstructive surgery (DBBMC, EDTA, CHX)                                                                 | DBBMC, EDTA 24%, CHX 1% gel, titanium                                                                                               | 5 years   | Uniform (all SLA surfaces)                          |

|                                                      |                                                      |                                                                                                 |       |                               |                                                                                                                 |                                                                                                                                                    |                                          |                                                                                     |
|------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------|-------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------|
|                                                      |                                                      |                                                                                                 |       |                               | gel, titanium<br>curettes,<br>connective<br>tissue graft)                                                       | curettes,<br>titanium<br>brush,<br>connective<br>tissue graft,<br>systemic<br>antibiotics                                                          |                                          |                                                                                     |
| <b>Romandini<br/>et al. (2024)<br/>(16)</b>          | Mixed<br>(Turned and<br>Modified)                    | Nobel<br>Biocare<br>(74.2%),<br>Astra Tech<br>(18%),<br>Straumann<br>(6.7%),<br>Neoss<br>(1.1%) | 149   | 267                           | Non-<br>reconstructi<br>ve surgical<br>therapy<br>(titanium-<br>coated<br>curettes,<br>systemic<br>antibiotics) | Titanium-<br>coated<br>curettes,<br>saline/CHX<br>gauze,<br>selective<br>systemic<br>antibiotics<br>(Amoxicillin)                                  | Mean 7<br>years<br>(range 1-18<br>years) | Modified<br>surfaces<br>significantly<br>higher<br>implant loss<br>risk<br>(HR=4.5) |
| <b>Roos-<br/>Jansäker<br/>et al. (2011)<br/>(29)</b> | Primarily<br>machined,<br>few<br>modified<br>(rough) | Brånemark<br>(majority),<br>Astra Tech<br>(minority)                                            | 38/32 | 65/56                         | Reconstruct<br>ive surgery<br>(Algipore ±<br>membrane)                                                          | Algipore,<br>Osseoquest<br>membrane,<br>H <sub>2</sub> O <sub>2</sub> 3%,<br>antibiotics<br>(amoxicillin,<br>metronidaz<br>ole), CHX<br>0.1% rinse | 3 years                                  | No<br>significant<br>impact<br>reported<br>(mostly<br>machined)                     |
| <b>Roos-<br/>Jansäker<br/>et al. (2014)<br/>(30)</b> | Primarily<br>machined,<br>few<br>modified<br>(rough) | Brånemark<br>(majority),<br>Astra Tech<br>(minority)                                            | 38/25 | Not initially<br>specified/45 | Reconstruct<br>ive surgery<br>(Algipore ±<br>membrane)                                                          | Algipore,<br>Osseoquest<br>membrane,<br>H <sub>2</sub> O <sub>2</sub> 3%,<br>antibiotics<br>(amoxicillin,<br>metronidaz<br>ole), CHX<br>rinse      | 5 years                                  | No<br>significant<br>impact<br>(mostly<br>machined<br>surfaces)                     |
| <b>Schwarz et<br/>al. (2009)<br/>(31)</b>            | Mixed<br>(machined<br>and rough<br>surfaces)         | Brånemark,<br>Camlog, ITI,<br>KSI Bauer<br>Schraube,<br>Zimmer, ZL-<br>Duraplant                | 22/19 | 22/19                         | Regenerativ<br>e surgery<br>(NBM+CM<br>vs. NHA)                                                                 | NBM<br>(BioOss),<br>CM<br>(BioGide),<br>NHA<br>(Ostim),<br>plastic<br>curettes,<br>CHX 0.2%                                                        | 48 months                                | Surface-<br>specific<br>outcomes<br>not detailed                                    |

**Table 2b: Comparative outcomes of clinical studies on peri-implantitis treatments: implant surfaces, secondary outcomes, and long-term surface impact**

| Author (Year)                          | Implant Loss (%)                                   | Disease Resolution                                                        | Radiographic Bone Loss/Gain                                   | Mean PD Post-treatment                                  | Subgroup Analysis (by surface & approach)                                               | Conclusions Surface Impact                                                                                                                                                    |
|----------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aghazadeh et al. (2022) (32)</b>    | AB: 1 fractured, BDX: 1 fractured (both initially) | PD reduction<br>AB: 1.7 mm, BDX: 2.8 mm at 5 years                        | AB: -0.7 mm (loss), BDX: +1.6 mm (gain)                       | AB: PD reduced by 1.7 mm, BDX: by 2.8 mm at 5 years     | BDX superior to AB; reconstructive approach only, no surface-specific outcomes detailed | BDX superior to AB; supportive maintenance every 3 months emphasized. No significant difference (turned vs medium-rough surfaces)                                             |
| <b>Carcuac et al. (2020) (15)</b>      | 20.8%                                              | Not explicitly combined; PD≥6mm at 1 year = increased recurrence (OR 7.4) | Stable from year 1; further loss >1mm in 13.1%                | 5.0 mm (stable from year 1 at 4.9 mm)                   | Non-modified: 17% recurrence, Modified: 52% recurrence; no explicit surgical subgroup   | Surgical treatment effective, but 44% recurrence; modified surfaces, deep residual PD major risk factors. Modified surfaces higher recurrence risk (OR 5.1, 95% CI: 1.6–16.5) |
| <b>Deppe et al. (2007) (18)</b>        | 17.8%                                              | Superior PD reduction laser (1.0-2.4 mm at 4 months)                      | Laser better short-term bone gain; long-term similar outcomes | Laser: 1.0-2.4 mm, Conventional: 2.4-7.9 mm (long-term) | Laser superior in soft tissue resection; similar in augmentation                        | Laser beneficial in non-reconstructive procedures; similar long-term results in augmentation. Superior outcomes with CO <sub>2</sub> laser in non-reconstructive therapy      |
| <b>Jemt &amp; Eriksson (2021) (19)</b> | 16.7%                                              | Not explicitly reported; primarily bone-level outcomes                    | Increased bone loss post-treatment (0.26 mm/year)             | PD not explicitly reported                              | No significant differences between surface types; increased bone loss post-treatment    | Surgical treatment ineffective long-term; bone loss accelerated post-treatment, edentulous worse prognosis. No significant difference (turned vs moderately                   |

|                                          |                        |                                                                                    |                                                                 |                                                                   |                                                                                                    |                                                                                                                                                           |
|------------------------------------------|------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          |                        |                                                                                    |                                                                 |                                                                   |                                                                                                    | rough surfaces)                                                                                                                                           |
| <b>Khoury &amp; Buchmann (2001) (20)</b> | 0% explicitly reported | PD improvement: Bone alone 8.0→2.9 mm, worse with membranes (7.7→5.1 mm)           | Significant bone gain (bone alone 3.2 mm, membranes lower)      | Bone alone: 2.9 mm; Non-resorbable: 2.8 mm; Bioabsorbable: 5.1 mm | Bone alone superior; membranes caused high complications                                           | Bone grafting alone effective; membranes did not enhance outcomes, increased complications. No surface comparison (all rough)                             |
| <b>La Monaca et al. (2024) (22)</b>      | 8.8%                   | Composite success: 53%, PD significantly reduced to 2.95 mm at 10 years            | Mean bone gain: 1.07 mm (stable 10 years)                       | Reduced from 6.33 mm to 2.95 mm at 10 years                       | Reconstructive approach only; stable bone gain, PD improvement, composite success 53%              | Long-term reconstructive therapy stable; supportive therapy crucial, defect severity not significant. All rough (TiUnite); no surface-specific comparison |
| <b>La Monaca et al. (2018) (21)</b>      | 0%                     | PD reduced initially, increased by 5 years (4.62 mm, nonsignificant from baseline) | Initial significant bone gain (1 year), gradual loss by 5 years | 4.62 mm at 5 years (from 5.93 mm baseline)                        | Reconstructive surgery initially effective, unstable outcomes long-term                            | Initial benefits lost over time, no implants lost; unpredictable outcomes long-term. All rough (TiUnite); no surface comparison                           |
| <b>Leonhardt et al. (2003) (23)</b>      | 27%                    | Significant reduction plaque (100%→11%) and bleeding (100%→5%)                     | Stable in 9 implants, gain in 6 implants, loss in 4 implants    | Not explicitly reported; clinical signs significantly improved    | Non-reconstructive approach; success 58%, significant clinical improvements                        | Limited success (58%); smoking negatively impacted outcomes. All rough surfaces; not directly analyzed                                                    |
| <b>Mercado et al. (2018) (24)</b>        | 0%                     | Mean PD from 8.9 mm to 3.5 mm; BOP/suppurative from 100% to 20%                    | Bone gain (6.92 mm to 2.60 mm mean bone loss at 36 months)      | Reduced from 8.9 mm to 3.5 mm at 36 months                        | Regenerative (reconstructive) approach only, significant PD reduction and bone gain; success 56.7% | Regenerative treatment effective (56.7% success); regular supportive therapy critical. Not specifically analyzed (micro-rough surfaces)                   |
| <b>Noelken et al. (2023) (25)</b>        | 8.3%                   | PD significantly reduced (5.05 to 3.08 mm),                                        | Significant bone gain (Interproximal 3.1 mm, Buccal 3.5         | Final PD 3.08 mm                                                  | LAPIDER effective for severe defects; significant                                                  | LAPIDER provided substantial regeneration and aesthetic                                                                                                   |

|                                         |                                      |                                                                                 |                                                                                     |                                                                    |                                                                                                        |                                                                                                                                                                              |
|-----------------------------------------|--------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         |                                      | BOP 100% to 36.4%                                                               | mm, Lingual 1.46 mm)                                                                |                                                                    | hard/soft tissue regeneration                                                                          | improvements. No direct comparison; rough surfaces only                                                                                                                      |
| <b>Roccuzzo et al. (2017) (26)</b>      | 16.7% overall (SLA:16.7%, TPS:28.6%) | Significant PD reduction, SLA (3.2 mm), TPS (3.4 mm); BOP SLA (7.5%), TPS (30%) | Significant bone fill SLA (2.1 mm gain), TPS (2.0 mm gain)                          | SLA: 3.2 mm, TPS: 3.4 mm                                           | Regenerative approach better for SLA vs. TPS implants                                                  | DBBMC effective; SLA surfaces significantly better clinical outcomes than TPS.                                                                                               |
| <b>Roccuzzo et al. (2020) (27)</b>      | SLA:20%, TPS:45%                     | Significant PD reduction SLA (3.2 mm), TPS (3.4 mm); BOP significantly reduced  | SLA substantial gain (2.7 mm), TPS moderate gain (2.0 mm)                           | SLA: 3.2 mm, TPS: 3.5 mm                                           | SLA implants superior long-term survival/outcomes                                                      | DBBMC stable outcomes; SLA significantly better than TPS long-term                                                                                                           |
| <b>Roccuzzo et al. (2021) (28)</b>      | 17%                                  | PD 6.89 mm to 4.06 mm; BOP 70.6% to 17.2%                                       | Substantial bone fill; no numeric specifics reported                                | Reduced from 6.89 mm to 4.06 mm at 5 years                         | Reconstructive approach only; SLA implants only, survival 80%, success 45.3%, significant PD reduction | Reconstructive protocol effective; adherence to supportive therapy significantly improves outcomes.                                                                          |
| <b>Romandini et al. (2024) (16)</b>     | 19.9%                                | Not explicitly; high recurrence, retreatment common (24.3%)                     | Mean additional bone loss: 0.97 mm; >1 mm loss in 42.4%                             | Not explicitly detailed post-treatment; baseline deepest PD 7.8 mm | Turned surfaces significantly better prognosis; modified surfaces high loss risk (HR=4.5)              | High recurrence; surface type crucial predictor, severe baseline bone loss/suppurative increase loss risk. Modified surfaces significantly higher implant loss risk (HR=4.5) |
| <b>Roos-Jansåker et al. (2011) (29)</b> | 0%                                   | Clinical measures not explicitly detailed                                       | Stable bone fill: bone graft (1.3 mm), membrane (1.6 mm), no significant difference | Not explicitly reported                                            | Stable outcomes, no significant membrane advantage                                                     | Aligipore ± membrane effective, stable bone fill, membrane no additional advantage. No significant impact reported (mostly machined)                                         |
| <b>Roos-Jansåker et al. (2014) (30)</b> | 0%                                   | PD reduction: 3.0-3.3 mm, BOP significantly reduced                             | Bone gain stable: 1.1-1.3 mm, no significant membrane advantage                     | 2.6-2.7 mm                                                         | No significant advantage with membrane                                                                 | Stable results; membrane no additional advantage over bone substitute alone.                                                                                                 |

|                                   |                                         |                                                                     |                                           |                             |                                      |                                                                                                          |
|-----------------------------------|-----------------------------------------|---------------------------------------------------------------------|-------------------------------------------|-----------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------|
|                                   |                                         |                                                                     |                                           |                             |                                      | No significant impact (mostly machined surfaces).                                                        |
| <b>Schwarz et al. (2009) (31)</b> | 0% explicitly reported (1 discontinued) | NBM+CM: PD reduction 2.5 mm, NHA: 1.1 mm, BOP significantly reduced | NBM+CM superior bone fill compared to NHA | NBM+CM: 4.6 mm, NHA: 5.8 mm | NBM+CM significantly better than NHA | NBM+CM superior long-term clinical/radiographic outcomes vs NHA. Surface-specific outcomes not detailed. |

*Combined Legend for Tables 2a and 2b*

AB, Autogenous Bone;  
 BDX, Bovine-derived Xenograft;  
 BOP, Bleeding on Probing;  
 CAL, Clinical Attachment Level;  
 CHX, Chlorhexidine;  
 CI, Confidence Interval;  
 CM, Collagen Membrane;  
 CO<sub>2</sub>, Carbon dioxide;  
 CT, Connective Tissue;  
 DBBMC, Deproteinized Bovine Bone Mineral with Collagen;  
 DH, Defect Height;  
 EMD, Enamel Matrix Derivative;  
 ePTFE, expanded Polytetrafluoroethylene;  
 H<sub>2</sub>O<sub>2</sub>, Hydrogen Peroxide;  
 HR, Hazard Ratio;  
 LAPIDER, Laser-Assisted Peri-Implant Defect Regeneration;  
 MBL, Marginal Bone Level;  
 MDBA, Mineralized Dehydrated Bone Allograft;  
 NBM, Natural Bone Mineral;  
 NHA, Nanocrystalline Hydroxyapatite;  
 OR, Odds Ratio;  
 PD, Probing Depth;  
 P-IS, Peri-implantitis Surgery;  
 SLA, Sandblasted Large grit Acid-etched;  
 SOP, Suppuration on Probing;  
 TPS, Titanium Plasma-Sprayed;  
 β-TCP, Beta-tricalcium phosphate.

### **Risk of bias**

The risk of bias assessment for the 17 included studies highlights several concerns across different domains.

The RCT (Schwarz 2009) had some concerns(31).

Among the non-randomized studies, serious risk of bias was frequently noted in selection of participants (e.g., Aghazadeh 2022, Carcuac 2020, La Monaca 2024) (15, 17, 22)and missing data (e.g., Roccuzzo 2020, Romandini 2024, Roos-Jansaker 2014) (16, 27, 30). However, intervention classification and reporting of results were generally at low risk across studies. Confounding and deviations from interventions were rated as moderate risk in most cases.

Ten studies were judged to have a serious overall bias, primarily due to participant selection and missing data. The remaining studies had a moderate overall bias, with issues mainly related to confounding and missing data.

Certainty of evidence for the main outcomes was assessed using the GRADE approach and is presented in Supplementary Table S1. Overall, the certainty was judged to be very low to low across all outcomes, primarily due to serious risk of bias, high heterogeneity of diagnostic criteria and outcome definitions, small sample sizes, and imprecision of effect estimates.

**Table 3a: Risk of bias assessment for RCTs: RoB 2 risk of bias assessment**

| Study             | Bias arising from the randomization process | Bias due to deviations from intended interventions | Bias due to missing outcome data | Bias in measurement of the outcome | Bias in selection of the reported result | Overall bias  |
|-------------------|---------------------------------------------|----------------------------------------------------|----------------------------------|------------------------------------|------------------------------------------|---------------|
| Schwarz 2009 (31) | Low                                         | Low                                                | Some concerns                    | Low                                | Low                                      | Some concerns |

**Table 3b: Risk of bias assessment for non- RCTs: ROBINS-i risk of bias assessment**

| Study                | Bias due to confounding | Bias in selection of participants into the study | Bias in classification of interventions | Bias due to deviations from intended interventions | Bias due to missing data | Bias in measurement of outcomes | Bias in selection of reported results | Overall bias |
|----------------------|-------------------------|--------------------------------------------------|-----------------------------------------|----------------------------------------------------|--------------------------|---------------------------------|---------------------------------------|--------------|
| Aghazadeh 2022 (32)  | Moderate                | Serious                                          | Low                                     | Moderate                                           | Serious                  | Moderate                        | Low                                   | Serious      |
| Carcuac 2020 (15)    | Moderate                | Serious                                          | Low                                     | Moderate                                           | High                     | Low                             | Low                                   | Serious      |
| Deppe 2007 (18)      | Moderate                | Moderate                                         | Low                                     | Moderate                                           | Serious                  | Moderate                        | Low                                   | Serious      |
| Jemt et al 2021 (19) | Moderate                | Serious                                          | Low                                     | Moderate                                           | Serious                  | Moderate                        | Low                                   | Serious      |
| Khoury 2001 (20)     | Serious                 | Low                                              | Moderate                                | Moderate                                           | Low                      | Moderate                        | Low                                   | Moderate     |
| La Monaca 2018 (21)  | Moderate                | Moderate                                         | Low                                     | Moderate                                           | Serious                  | Low                             | Low                                   | Moderate     |
| La Monaca 2024 (22)  | Serious                 | Serious                                          | Low                                     | Moderate                                           | Serious                  | Moderate                        | Low                                   | Serious      |
| Leonhardt 2003 (23)  | Moderate                | Low                                              | Low                                     | Moderate                                           | Low                      | Moderate                        | Low                                   | Moderate     |
| Mercado 2018 (24)    | Moderate                | Low                                              | Low                                     | Moderate                                           | Low                      | Low                             | Low                                   | Moderate     |
| Noelken 2023 (25)    | Moderate                | Moderate                                         | Low                                     | Moderate                                           | Serious                  | Moderate                        | Low                                   | Moderate     |
| Roccuzzo 2017 (26)   | Moderate                | Moderate                                         | Low                                     | Moderate                                           | Moderate                 | Low                             | Low                                   | Moderate     |

|                                |          |         |     |          |         |          |     |         |
|--------------------------------|----------|---------|-----|----------|---------|----------|-----|---------|
| <b>Roccuzzo 2020 (27)</b>      | Moderate | Serious | Low | Moderate | Serious | Low      | Low | Serious |
| <b>Roccuzzo 2021 (28)</b>      | Moderate | Serious | Low | Moderate | Serious | Moderate | Low | Serious |
| <b>Romandini 2024 (16)</b>     | Moderate | Serious | Low | Moderate | Serious | Moderate | Low | Serious |
| <b>Roos-Jansaker 2011 (29)</b> | Moderate | Serious | Low | Moderate | Serious | Low      | Low | Serious |
| <b>Roos-Jansaker 2014 (30)</b> | Moderate | Serious | Low | Moderate | Serious | Low      | Low | Serious |

## 4. Discussion

This systematic review aimed to evaluate the impact of implant surface characteristics on the long-term outcomes of surgical treatment of peri-implantitis. The main findings indicate that modified (rough) surfaces are consistently associated with higher recurrence and implant loss compared with turned (machined) surfaces. Within rough surfaces, SLA implants achieved more favorable outcomes than TPS, particularly in reconstructive contexts (15, 16, 26-28). In contrast, smooth implants demonstrated comparatively lower recurrence (23). These results underscore that implant surface topography can be a determinant of surgical treatment prognosis.

The effectiveness of peri-implantitis surgery is influenced by both treatment modality and implant surface. For non-regenerative procedures, modified surfaces were repeatedly associated with worse outcomes: Carcuac et al. (2020) reported a 44% recurrence rate with rough implants (15), while Romandini et al. (2023) identified modified surfaces such as TiUnite and SLA as predictors of implant loss (16). In contrast, turned surfaces showed more stable disease suppression (23).

For regenerative approaches, implant surface also played a key role. SLA implants demonstrated favorable long-term bone gain and PD reduction (26-28), whereas TPS implants performed poorly even when grafting was applied (27). Although smooth implants were less frequently studied in regenerative contexts, available evidence suggests they may perform adequately when combined with supportive therapy (29, 30).

Bone regeneration outcomes differed substantially according to surface characteristics. Greater bone fill was generally reported around rough surfaces when grafting materials were used. Khoury and Buchmann (2001) observed a 2.4 mm gain at 12 months using autogenous grafts on rough implants (20). Roccuzzo et al. (2017, 2020) reported significant defect reduction with xenografts, particularly in SLA implants, while TPS implants showed limited stability (26, 27). Comparable results with alloplastic materials were also noted (33, 34). However, bone regeneration around smooth surfaces was less favorable: Roos-Jansåker et al. (2014) reported limited improvement with alloplastic grafts (30). Thus, while rough implants may predispose to recurrence, they also appear to support more pronounced bone regeneration after reconstructive procedures.

This paradox may be explained by surface-related biology. Rough surfaces are harder to decontaminate and accumulate more plaque (35, 36), yet they may stabilize the

coagulum and promote defect fill (37). Accordingly, radiographic bone gain does not necessarily correspond to re-osseointegration, as several animal studies identified connective tissue interposition rather than true reattachment (38-40).

The role of membranes in guided bone regeneration (GBR) has also been linked to implant surfaces. Khoury et al. (2001) showed greater bone gain with non-resorbable membranes around rough implants (20), while Deppe et al. (2007) observed comparable results with resorbable membranes (18). These data suggest that both membrane type and surface roughness influence regenerative outcomes. Furthermore, clinical studies and experimental models in dogs indicate that rough surfaces generally achieve greater defect fill than smooth surfaces under GBR conditions (37).

Surface characteristics may also impact soft tissue attachment. Excessively smooth surfaces can impair mucosal adhesion, as Quirynen et al. (1996) observed attachment loss on polished abutments compared with stable CAL around commercially available surfaces (41). Other studies support that maintaining a certain degree of roughness enhances soft tissue sealing (42). These findings provide a biological explanation for the improved clinical outcomes of rough implants after GBR, despite their higher susceptibility to recurrence.

Interpretation of the evidence is complicated by considerable heterogeneity. Defect morphology influences outcomes, with narrower defects showing better results (17, 43), yet most studies failed to provide detailed descriptions, limiting cross-study comparisons. Moreover, peri-implantitis definitions varied widely: Rocuzzo et al. (2017) required  $\geq 6$  mm PD and bone loss exceeding three implant threads (26), while Mercado et al. (2018) used  $\geq 4$  mm PD and  $\geq 20\%$  radiographic bone loss (24). Measurement variability further complicates interpretation (44). Such inconsistencies directly affect assessment of surface-related outcomes and hinder robust comparisons across studies.

This review is limited by the substantial heterogeneity among the included studies, particularly in peri-implantitis diagnostic criteria, defect morphology, surgical techniques, and outcome measures. Most studies were small in size, lacked standardized definitions, and many were judged to have a serious overall risk of bias, especially in participant selection and missing data. Confounding variables were insufficiently controlled, further reducing certainty. Furthermore, the restriction to English-language studies may have introduced language bias, potentially leading to omission of relevant non-English publications. Applying the GRADE framework, the certainty of the available evidence was rated as very low to low for all main outcomes, reflecting methodological shortcomings and heterogeneity among the included studies. These limitations restrict the generalizability of the findings and reinforce the need for well-designed, adequately powered randomized controlled trials with standardized definitions and longer follow-up.

## **Conclusion**

The effectiveness of peri-implantitis surgery is influenced by implant surface characteristics and treatment modality. Modified surfaces are generally more prone to recurrence and implant loss, with SLA implants performing better than TPS, while turned surfaces appear less susceptible but remain insufficiently studied in regenerative

contexts. Reconstructive approaches combined with supportive care consistently provide the most favorable outcomes. Given the very low to low certainty of the evidence with heterogenous results, current findings should be interpreted with caution, and well-designed long-term randomized trials with standardized definitions and consistent surface classifications are urgently needed. Future trials should adopt standardized outcome definitions (e.g., PD thresholds, BOP, radiographic bone loss criteria) to allow comparability across studies. Research should focus on RCTs directly comparing surface types, long-term follow-up, and adjustment for confounding factors such as defect morphology and maintenance compliance. Addressing these gaps will clarify the role of implant surface modifications.

### **Clinical Implications:**

When planning peri-implantitis surgery, implant surface characteristics should be taken into account, but they must not be considered in isolation. Evidence indicates that reconstructive approaches yield more reliable outcomes than non-reconstructive ones, particularly for rough implants, with SLA surfaces performing more favorably than TPS. Turned (machined) surfaces appear less prone to recurrence, although data on regenerative protocols remain scarce. These observations suggest that implant surface may influence prognosis, yet it represents only one part of a complex clinical picture. Patient-related risk factors (such as smoking, systemic conditions, low compliance and/or adherence to supportive care) exert a profound effect on long-term success and may outweigh surface-related differences. Surgical decision-making should therefore be individualized, integrating implant surface type, defect morphology, patient risk profile, and anticipated compliance. The use of biomaterials and barrier membranes may enhance regenerative outcomes around rough implants, but clinicians should be cautious, as radiographic bone gain does not necessarily reflect true re-osseointegration, and complete defect resolution is rarely achievable. Nevertheless, these clinical implications must be interpreted with caution. The available evidence is heterogeneous, often based on small studies with differing peri-implantitis definitions, inconsistent outcome measures, and a serious overall risk of bias. The evidence was rated as very low to low for all main outcomes. This means that while current data can guide clinical choices, they cannot provide definitive recommendations.

## 5. Apendix

### Appendix 1 –excluded studies

| Authors and Date                 | Study Title                                                                                                                                                               | Reason for Exclusion                                                                                                       |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Afrashtehfar et al. 2024 (45)    | Guided bone regeneration improves defect fill and reconstructive outcomes in 3-wall peri-implantitis defects                                                              | Implant type not taken into consideration                                                                                  |
| Astolfi et al. 2021 (46)         | Influence of removing or leaving the prosthesis after regenerative surgery in peri-implant defects: retrospective study: 32 clinical cases with 2 to 8 years of follow-up | Outcomes reported not correlated to implant type                                                                           |
| Behneke et al. 2000 (47)         | Treatment of peri-implantitis defects with autogenous bone grafts: six-month to 3-year results of a prospective study in 17 patients                                      | Implant type not taken into consideration                                                                                  |
| Berglundh 2018 (48)              | Long- term outcome of surgical treatment of periimplantitis. A 2-11-year retrospective study                                                                              | Minimum study duration less than 3 years                                                                                   |
| Bianchini et al. 2020 (49)       | Implantoplasty enhancing peri-implant bone stability over a 3-year follow-up: a case series                                                                               | Implant type not taken into consideration                                                                                  |
| Bianchini et al. (2024) (50)     | Clinical and radiographic outcomes of resective surgery with adjunctive implantoplasty over a 6- to 11-year follow-up: a case series                                      | Implants treated with implantoplasty, which creates a modified surface texture differing from the original implant surface |
| Carcuac et al. 2016 (51)         | Adjunctive systemic and local antimicrobial therapy in the surgical treatment of peri-implantitis: a randomized controlled clinical trial                                 | Study duration less than 3 years                                                                                           |
| Chiang et al. 2024 (52)          | Operating microscope-assisted reconstructive strategy for peri-implantitis: A case series report                                                                          | Implant type not taken into consideration                                                                                  |
| Cortellini et al. 2021 (53)      | Papilla preservation and minimally invasive surgery for the treatment of peri-implant osseous defects. Clinical and radiographic outcomes of a 5-year retrospective study | Implant type not taken into consideration                                                                                  |
| Froum et al. 2012 (54)           | Successful management of peri-implantitis with a regenerative approach: a consecutive series of 51 treated implants with 3- to 7.5-year follow-up                         | Outcomes mentioned not correlated to implant surfaces                                                                      |
| Froum et al. 2015 (55)           | A regenerative approach to the successful treatment of peri-implantitis: a consecutive series of 170 implants in 100 patients with 2- to 10-year follow-up                | Implant type not taken into consideration                                                                                  |
| Khayat et al. 2024 (56)          | Bone regeneration following implantoplasty: a retrospective cohort study with long-term radiographic assessment                                                           | Implant type not taken into consideration                                                                                  |
| Lombardo et al. 2019 (57)        | Successful management of peri-implantitis around short and ultrashort single-crown implants: a case series with a 3-year follow-up                                        | Implant type not taken into consideration                                                                                  |
| Monje et al. 2022 (58)           | Principles of combined surgical therapy for the management of peri-implantitis                                                                                            | Incorrect study design                                                                                                     |
| Parma-Benfenati et al. 2020 (59) | Long-term outcome of surgical regenerative treatment of peri-implantitis: a 2- to 21-year retrospective evaluation                                                        | Study duration less than 3 years (varied for 2 to 21 years)                                                                |
| Renvert et al. 2012 (60)         | Surgical therapy for the control of peri-implantitis                                                                                                                      | Incorrect study design                                                                                                     |
| Renvert et al 2024 (61)          | The efficacy of reconstructive therapy in the surgical management of peri-implantitis: A 3-year follow-up of a randomized clinical trial                                  | Implant type not taken into consideration                                                                                  |
| Sarmiento et al. 2018 (62)       | Surgical alternatives for treating peri-implantitis                                                                                                                       | Implant type not taken into consideration                                                                                  |
| Schwarz et al. 2015 (63)         | Reentry after combined surgical resective and regenerative therapy of advanced peri-implantitis: a retrospective analysis of five cases                                   | Implant type not taken into consideration                                                                                  |
| Schwarz et al. 2014 (64)         | Combined surgical therapy of advanced peri-implantitis lesions with concomitant soft tissue volume augmentation. A case series                                            | Study duration less than 3 years                                                                                           |

|                                 |                                                                                                                                          |                                                                                                                            |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Schwarz et al. 2013 (65)</b> | Four-year follow-up of combined surgical therapy of advanced peri-implantitis evaluating two methods of surface decontamination          | Implants treated with implantoplasty, which creates a modified surface texture differing from the original implant surface |
| <b>Schwarz et al. 2017 (66)</b> | Combined surgical therapy of advanced peri-implantitis evaluating two methods of surface decontamination: a 7-year follow-up observation | Implants treated with implantoplasty, which creates a modified surface texture differing from the original implant surface |
| <b>Wang et al. 2021 (67)</b>    | Laser-assisted regenerative surgical therapy for peri-implantitis: A randomized controlled clinical trial                                | Study duration less than 3 years                                                                                           |

## Appendix 2 – Full Search Strategies

| Database                    | Search Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PubMed (MEDLINE via PubMed) | (Periimplantitis OR peri-implantitis OR peri implantitis OR periimplant OR peri-implant OR peri implant)<br>AND<br>(treatment outcome OR therapy OR surgical treatment OR regenerative OR regeneration OR tissue regeneration OR reconstructive surgery OR bone graft OR bone substitute OR membranes OR surgical flap OR open flap debridement OR resective OR implantoplasty OR surface decontamination)<br>AND<br>(surface characteristics OR surface roughness OR material characteristics OR titanium surface OR implant types OR implant surfaces OR surface topography OR surface analysis)<br>AND<br>(implant survival OR bone loss OR recurrence OR retreatment OR radiographic stability OR long-term OR 3 years OR follow-up)                                                                                                                                             |
| Embase                      | ('periimplantitis'/exp OR periimplantitis OR 'peri-implantitis' OR 'peri implantitis' OR periimplant OR 'peri-implant' OR 'peri implant')<br>AND<br>(('treatment outcome'/exp OR therapy OR 'surgical treatment'/exp OR 'regenerative therapy'/exp OR regeneration OR 'tissue regeneration' OR 'reconstructive surgery'/exp OR 'bone graft'/exp OR 'bone substitute'/exp OR membranes OR 'surgical flap' OR 'open flap debridement' OR resective OR implantoplasty OR 'surface decontamination')<br>AND<br>(('surface property'/exp OR 'surface roughness'/exp OR 'material property'/exp OR 'titanium surface' OR 'implant type'/exp OR 'implant surface'/exp OR 'surface topography'/exp OR 'surface analysis')<br>AND<br>(('dental implant survival'/exp OR 'bone loss'/exp OR recurrence OR retreatment OR 'radiographic stability' OR 'long term' OR '3 years' OR 'follow-up')) |
| Cochrane Library            | (periimplantitis OR "peri-implantitis" OR "peri implantitis" OR periimplant OR "peri-implant" OR "peri implant")<br>AND<br>("treatment outcome" OR therapy OR "surgical treatment" OR regenerative OR regeneration OR "tissue regeneration" OR "reconstructive surgery" OR "bone graft" OR "bone substitute" OR membranes OR "surgical flap" OR "open flap debridement" OR resective OR implantoplasty OR "surface decontamination")<br>AND<br>("surface characteristics" OR "surface roughness" OR "material characteristics" OR "titanium surface" OR "implant types" OR "implant surfaces" OR "surface topography" OR "surface analysis")<br>AND<br>(("implant survival" OR "bone loss" OR recurrence OR retreatment OR "radiographic stability" OR "long term" OR "3 years" OR "follow-up"))                                                                                     |

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### **Conflict of Interest**

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. The authors declare that no financial support was received from implant manufacturers or commercial entities.

### **Author Contributions**

Alkisti Zekeridou (AZ) and Panagiotis Gardelis (PG) contributed equally to the conception, design, and drafting of the systematic review protocol, as well as the data acquisition and interpretation. Catherine Giannopoulou (CG) provided critical revisions and contributed to the methodological framework and data interpretation. Andreas Stavropoulos (AS) supervised the project, provided expert guidance, contributed to the conception, design of the systematic review protocol and contributed to the final review and approval of the manuscript. All authors (AZ, PG, CG, AS) have read and approved the final version of the manuscript.

## DISCUSSION

Le traitement chirurgical de la péri-implantite est aujourd'hui reconnu comme une modalité thérapeutique efficace permettant d'améliorer les paramètres cliniques et radiographiques associés à cette pathologie inflammatoire complexe. Plusieurs revues systématiques ont rapporté des améliorations significatives des profondeurs de poche, du saignement au sondage et des niveaux osseux péri-implantaires après différentes approches chirurgicales, qu'elles soient non régénératives ou reconstructives (47, 69, 70). Dans ce contexte, les procédures régénératives ont été développées avec l'objectif biologique explicite d'obtenir une ré-osseointégration, et ont montré des degrés variables de succès selon les protocoles appliqués (47, 71, 72).

Il est désormais bien établi que la ré-osseointégration peut survenir sur une surface implantaire précédemment contaminée, pour autant qu'une décontamination adéquate soit réalisée (70). Cette capacité biologique a été démontrée dans plusieurs études animales utilisant des modèles de péri-implantite induite chez le chien (73-75). Bien que l'extrapolation directe de ces résultats à la pratique clinique humaine puisse être discutée, le modèle canin de péri-implantite induite par ligature est largement considéré comme une représentation acceptable des défauts péri-implantaires observés chez l'humain et demeure couramment utilisé dans la recherche expérimentale sur le traitement de la péri-implantite (76).

### **Influence de la morphologie du défaut péri-implantaire**

La configuration du défaut osseux péri-implantaire constitue un facteur déterminant du pronostic thérapeutique. Des études expérimentales ont montré que des défauts plus étroits et mieux contenus présentaient un potentiel régénératif supérieur après traitement chirurgical de la péri-implantite induite (77). En parallèle, des études cliniques ont également souligné l'impact de la morphologie du défaut sur les résultats des traitements reconstructifs (77, 78).

Toutefois, une limite méthodologique majeure de la majorité des études cliniques incluses réside dans l'absence fréquente de description détaillée de la morphologie des défauts péri-implantaires, ce qui complique la comparaison inter-études des résultats radiographiques et limite l'interprétation du potentiel régénératif réel des différentes approches thérapeutiques.

### **Hétérogénéité des définitions de la péri-implantite**

L'interprétation des données disponibles est entravée par une hétérogénéité marquée dans les définitions de la péri-implantite. Bien que toutes les études incluses aient requis la présence d'une perte osseuse péri-implantaire, des différences substantielles ont été observées concernant les seuils diagnostiques de profondeur de sondage, de saignement, de suppuration au sondage, ainsi que la quantité minimale de perte osseuse requise.

Par exemple, Roos-Jansåker et al. (2011) ont défini la péri-implantite comme une perte osseuse progressive  $\geq$  trois spires associées à un saignement et/ou une suppuration (66), tandis que Schwarz et al. (2017) ont exigé la présence d'un défaut intra-osseux avec une PPD  $> 6$  mm et une composante intra-osseuse  $> 3$  mm radiographiquement détectable (79). Cette variabilité est d'autant plus problématique que les mesures des niveaux osseux péri-implantaires sont associées à une erreur inter-examineur estimée à environ 0,4 mm (80), ce qui peut influencer de manière significative la classification de la présence ou de l'absence de péri-implantite.

## **Rôle des caractéristiques de surface implantaire**

Les caractéristiques de surface implantaire ont été largement étudiées en relation avec différents aspects de la péri-implantite. Bien que certaines analyses n'aient pas identifié la surface implantaire comme un facteur prédictif direct de l'apparition de la maladie (81), il est clairement démontré que la topographie de surface influence de manière significative la formation et la maturation du biofilm bactérien (82), les implants rugueux présentant une accumulation et une rétention bactériennes plus marquées que les implants lisses (83).

Les données issues de la présente revue systématique indiquent que les surfaces modifiées sont globalement associées à des taux plus élevés de récurrence et de perte implantaire par rapport aux surfaces tournées. Carcuac et al. (2020) ont rapporté un taux de récurrence de 44 % après chirurgie de la péri-implantite autour d'implants à surface rugueuse (30), tandis que Romandini et al. (2024) ont identifié les surfaces modifiées, notamment TiUnite et SLA, comme des facteurs pronostiques indépendants de perte implantaire à long terme (32). À l'inverse, les implants à surface tournée ont montré une suppression plus stable de la maladie après traitement non régénératif (60).

Par ailleurs, une revue systématique de Renvert et al. (2012) a conclu que la capacité de ré-osseointégration pouvait être influencée par les caractéristiques de surface implantaire (51). D'un point de vue théorique, les surfaces rugueuses constituent un défi accru en termes de décontamination, ce qui pourrait compromettre le contrôle à long terme de l'inflammation. Toutefois, des études expérimentales ont montré une ré-osseointégration plus prononcée sur des surfaces rugueuses que sur des surfaces lisses après traitement chirurgical non régénératif (84). Persson et al. (2001) ont suggéré que les surfaces rugueuses pourraient offrir un support mécanique favorable à la stabilisation du caillot sanguin, facilitant ainsi la cicatrisation osseuse le long de la surface implantaire (84).

## **Résultats des traitements non régénératifs**

Les données cliniques disponibles ne permettent pas de mettre en évidence de différences nettes en termes de régénération osseuse entre les surfaces implantaires après chirurgie non régénérative. Leonhardt et al. (2003) ont rapporté à la fois des gains et des pertes osseuses autour d'implants à surface lisse après 12 mois de suivi, tandis que des résultats comparables ont été observés autour d'implants rugueux dans d'autres études (60).

Bien que la chirurgie résective, avec ou sans implantoplastie ou ostéoplastie, puisse être efficace dans le contrôle de la péri-implantite (51, 85), aucune différence radiographique claire liée à la topographie de surface implantaire n'a pu être identifiée.

### **Résultats des approches reconstructives et régénératives**

Lorsque le traitement chirurgical inclut l'utilisation de matériaux de comblement, une augmentation plus marquée du niveau osseux péri-implantaire est observée autour des implants rugueux. Le gain osseux le plus prononcé (2,4 mm) a été rapporté par Khoury et Buchmann (2001) après utilisation d'une greffe autogène (57). Des résultats similaires ont été observés avec des xélogreffes, en particulier autour des implants SLA, avec une stabilité à long terme rapportée jusqu'à 10 ans (63-65). En revanche, les implants TPS ont montré une stabilité osseuse limitée, même en présence de matériaux de comblement (63).

Des matériaux alloplastiques ont également montré une capacité de comblement des défauts péri-implantaires autour des implants rugueux (86, 87). En revanche, lorsque des matériaux alloplastiques ont été utilisés autour d'implants à surface lisse, l'effet sur le niveau osseux était nettement moins prononcé (67).

### **Régénération osseuse guidée, membranes et tissus mous**

La régénération osseuse guidée, combinant matériaux de comblement et membranes, semble induire globalement des gains osseux plus importants autour des implants rugueux que lisses, ce qui est en accord avec les résultats d'études animales après traitements non régénératifs (84). Toutefois, plusieurs études expérimentales ont mis en évidence la présence d'un tissu conjonctif interposé entre l'os néoformé et la surface implantaire (73, 74, 88), suggérant que la régénération osseuse radiographique ne correspond pas nécessairement à une véritable ré-osseointégration.

Le rôle des membranes demeure controversé. Khoury et Buchmann (2001) ont montré des gains osseux supérieurs avec des membranes non résorbables par rapport aux membranes résorbables autour d'implants rugueux (57), tandis que Deppe et al. (2007) ont rapporté des résultats comparables avec des membranes résorbables (55). Lorsque des membranes non résorbables ont été utilisées en association avec des matériaux alloplastiques autour d'implants lisses, les gains osseux étaient nettement inférieurs (66, 89).

Les caractéristiques de surface implantaire influencent également la qualité de l'attache des tissus mous péri-implantaires. Une rugosité excessive ou insuffisante peut compromettre le scellement muqueux. Quirynen et al. (1996) ont observé une perte d'attache autour de surfaces polies, tandis que des surfaces commerciales standard permettaient un maintien plus stable des niveaux d'attache (90). Ces observations sont soutenues par des études in vitro montrant une meilleure adhésion cellulaire sur des surfaces présentant un certain degré de rugosité (91).

### **Conclusion**

Les modifications de surface implantaire ont démontré leur capacité à améliorer l'osseointégration initiale et les taux de survie implantaire (6, 92). Toutefois, elles sont également associées à une susceptibilité accrue à l'accumulation bactérienne, ce qui pourrait contribuer au développement et à la récurrence des maladies péri-implantaires. Renvert et al. (2012) ont conclu que les preuves disponibles restaient limitées quant à l'influence directe des caractéristiques de surface sur la progression de la péri-implantite établie (51). Dans un contexte de prévalence croissante de la péri-implantite (9), l'identification de stratégies thérapeutiques optimales demeure une priorité clinique et scientifique.

Les résultats de ce travail suggèrent que l'efficacité du traitement chirurgical de la péri-implantite dépend à la fois des caractéristiques de surface implantaire et de la modalité thérapeutique choisie. Les surfaces modifiées sont globalement associées à un risque plus élevé de récurrence et de perte implantaire, bien que les implants à surface SLA présentent des résultats plus favorables que ceux à surface TPS. À l'inverse, les implants à surface tournée semblent moins sujets à la récurrence, mais leur comportement dans des contextes reconstructifs demeure insuffisamment documenté. Les approches reconstructives, lorsqu'elles sont associées à un programme de maintenance rigoureux, apparaissent comme les stratégies offrant les résultats les plus stables.

Toutefois, ces observations doivent être interprétées avec prudence en raison du faible niveau de certitude des preuves disponibles et de l'hétérogénéité méthodologique des études incluses. Des essais cliniques randomisés, bien conçus, avec des définitions standardisées des critères cliniques et radiographiques et des suivis à long terme, sont nécessaires pour mieux comprendre le rôle spécifique des surfaces implantaires et permettre des comparaisons fiables entre études.

En pratique, les caractéristiques de surface implantaire doivent être prises en compte dans la planification du traitement chirurgical de la péri-implantite, sans être considérées isolément.

La prise de décision clinique doit rester individualisée et intégrer les facteurs liés au patient, tels que le tabagisme, les conditions systémiques et la compliance au suivi, qui peuvent avoir un impact majeur sur le succès à long terme. Enfin, bien que l'utilisation de biomatériaux et de membranes puisse améliorer les résultats régénératifs, les cliniciens doivent garder à l'esprit que les gains osseux radiographiques ne reflètent pas nécessairement une ré-osseointégration complète et que la résolution totale des défauts péri-implantaires demeure rarement atteignable.

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# Impact of implant surface modifications on long-term outcome of surgical peri-implantitis treatment: a systematic review

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**Introduction:** Peri-implantitis is an inflammatory disease that compromises peri-implant tissues and supporting bone, potentially leading to implant loss. Although several surgical treatment strategies have been proposed, it remains unclear whether implant surface characteristics (smooth vs. rough) influence long-term treatment outcomes.

**Methods:** A systematic review was conducted to evaluate clinical studies with a minimum follow-up of 3 years that assessed the outcomes of surgical treatment of peri-implantitis in relation to implant surface type. Data extraction focused on recurrence of peri-implantitis, implant survival, clinical parameters, radiographic outcomes, and the type of surgical approach used (reconstructive vs. non-reconstructive).

**Results:** Seventeen clinical studies were included. Outcomes varied according to implant surface characteristics. Rough (modified) surfaces were generally associated with a higher risk of recurrence of peri-implantitis and implant loss compared with smooth (machined/turned) surfaces. Reconstructive surgical approaches, especially those involving bone grafts and membranes, demonstrated more favorable outcomes compared with non-reconstructive approaches.

**Discussion:** Despite observed trends, the certainty of the evidence remains low due to heterogeneity between studies, small sample sizes, and methodological limitations. Further well-designed long-term clinical trials are needed to clarify the role of implant surface characteristics in the long-term success of peri-implantitis surgical treatment.

**Systematic Review Registration:** PROSPERO (CRD420251129791).

## KEYWORDS

peri-Implantitis, surgical peri-implantitis treatment, treatment outcome, explantation, implant surface, long-term outcomes, bone loss, implant survival

## 1 Introduction

Dental implants have significantly advanced oral rehabilitation, providing highly predictable solutions for tooth replacement. For instance, a recent systematic review reported that long-term prospective studies on dental implants show high survival rates, typically exceeding 90% over 5–10 years and remaining around 78% after imputation at 20 years follow-up. In addition, five retrospective studies with  $\geq 20$  years of follow-up reported an implant survival rate of approximately 88%, including multifactorial causes (1).

However, despite the high survival rates, biological complications at implants are rather common. In particular, peri-implantitis, which is characterized by peri-implant mucosal inflammation and progressive bone loss, affects approximately 19.53% of patients and 12.53% of implants, highlighting its relevance in clinical practice (2). As the main etiological factor for peri-implantitis is the oral biofilm, microbial to implant surface interactions seem to play an important role in disease pathogenesis. Indeed, surface modifications (e.g., sandblasting, acid-etching, anodization, etc) aiming in enhancing implant surface bioactivity, substantially impact on microbial colonization and biofilm development (3–7). Indeed, although the incidence of peri-implantitis seems not to differ between modified and non-modified (i.e., turned) implants in the clinic, progression and severity of peri-implantitis appear linked to implant surface properties; specifically, pre-clinical *in vivo* studies indicate a faster disease progression at modified implants compared with turned implants, as well as differences in disease progression among various modified surfaces (5, 7). Moreover it seems that implant surface characteristics may impact on treatment outcomes both in the short-term but also on the long-term, with implants with a modified surface demonstrating less positive results and higher recurrence rates (3, 8, 9).

Despite technological advancements and improved treatment approaches, the impact of implant surface modifications on peri-implantitis outcomes remains unclear. Therefore, this systematic review aims to evaluate whether varying implant surface topographies influence clinical and radiographic outcomes following surgical peri-implantitis treatment in humans. The findings may offer critical insights guiding the selection of implant surface characteristics to enhance treatment efficacy.

## 2 Materials and methods

### 2.1 Study design

This review was performed following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and was registered in PROSPERO (ID: CRD420251129791).

### 2.2 Search strategy

To identify relevant studies, we systematically searched PubMed, Embase, and the Cochrane Library. The search strategy was carried out in English language from database inception for

articles published between 2000 and 2025. Two investigators (AZ and PG) independently reviewed the search results and screened the titles and abstracts. Full texts of all potentially eligible studies were obtained. In PubMed, the following search strategy was used: “(Periimplantitis OR peri-implantitis OR peri implantitis OR periimplant OR peri-implant OR peri implant) AND (treatment outcome OR therapy OR surgical treatment OR regenerative OR regeneration OR tissue regeneration OR reconstructive surgery OR bone graft OR bone substitute OR membranes OR surgical flap OR open flap debridement OR resective OR implantoplasty OR surface decontamination) AND (surface characteristics OR surface roughness OR material characteristics OR titanium surface OR implant types OR implant surfaces OR surface topography OR surface analysis) AND (implant survival OR bone loss OR recurrence OR retreatment OR radiographic stability OR long-term OR 3 years OR follow-up).” This search strategy was adapted to suit the other electronic sources. The reference lists of retrieved articles were also checked to identify additional studies of interest. Any inconsistencies were resolved by consensus with a third investigator (CG). The complete search strategies for all databases are provided in [Appendix 2](#).

### 2.3 Criteria for considering studies for this review

#### 2.3.1 Study design

Randomized controlled trials, prospective studies, retrospective studies, case-control studies, and case series were included. No specific cut-off criteria for sample size were applied, given the limited availability of data. Additionally, two case series with very small sample sizes were included due to their clinical relevance. Eligibility required that included studies explicitly reported the implant surface type(s) of the implants investigated.

#### 2.3.2 Population

Human studies. Patients with osseointegrated dental implants diagnosed with peri-implantitis, treated surgically, with a follow-up period of at least 3 years (or an average  $\geq 3$  years).

#### 2.3.3 Intervention

Surgical therapy for peri-implantitis.

#### 2.3.4 Comparator

Different implant surface types, characterized by variations in macro-, micro-, and nano-scale surface roughness, topography, and material composition. Surfaces were categorized as non-modified (i.e. turned, smooth, machined), modified (rough), or mixed (hybrid), depending on their reported surface characteristics.

### 2.4 Outcomes

#### 2.4.1 Primary outcome

- Percentage of implants with recurrence of peri-implantitis requiring re-treatment or explantation or simply defined as treatment failure by the authors.

## 2.4.2 Secondary outcomes

- Implant loss (due to any reason)
- Disease resolution defined by reduction of probing depth (PD) without bleeding on probing (BOP) or suppuration
- Radiographic bone loss or gain assessed by mean changes in bone levels or percentage of implants with stable bone levels post-treatment
- Mean probing depth (PD) post-treatment

Subgroup synthesis: The outcomes were further stratified based on implant surface types and surgical approach:

1. Turned (machined/non-modified surfaces)
  - a. Non-reconstructive surgical approach
  - b. Reconstructive surgical approach (regardless the technique or materials used)
2. Modified (rough surfaces)
  - a. Non-reconstructive surgical approach
  - b. Reconstructive surgical approach (regardless the technique or materials used)
3. Mixed or unspecified surfaces
  - a. Non-reconstructive surgical approach
  - b. Reconstructive surgical approach (regardless the technique or materials used)

## 2.5 Data collection

Two investigators independently extracted key data from the included articles. The inter-rater agreement for study selection was assessed using Cohen's kappa statistics. Inter-rater reliability was assessed using Cohen's kappa statistic on a subset of 20% of studies, yielding a kappa of 0.85, indicating a high level of agreement. Discrepancies were resolved through discussion or consultation with a third reviewer (CG). For each article, we extracted study features (i.e., study design, year of publication, number of enrolled patients), type of intervention, and outcome measures. Correct data extraction was controlled in a subset of randomly selected studies by the third investigator.

## 2.6 Assessment of risk of bias

Two investigators independently appraised the risk of bias of the included studies using the Cochrane Risk of Bias Tool 2.0 (RoB2) for RCTs. For non-RCTs the Risk of Bias in Non-randomized Studies of Interventions (ROBINS-I) tool was used. Any inconsistencies were resolved by consensus with a third investigator (CG).

## 2.7 Data synthesis

Preliminary analyses of available data revealed high heterogeneity, precluding meaningful meta-analysis. Therefore, a

narrative synthesis was conducted. These limitations included significant heterogeneity in implant surface types, surgical techniques, and reported outcome measures across studies. To enhance clarity and readability, findings were systematically summarized in tables according to pre-defined outcomes and subgroup analyses.

Data extraction was performed separately for each treatment group within studies containing multiple groups, while data from studies with a single treatment group were extracted accordingly. Results were categorized based on implant surface types and surgical approaches as follows:

1. Turned (machined/non-modified surfaces)
  - a. Non-reconstructive surgical approach
  - b. Reconstructive surgical approach (regardless the technique or materials used)
2. Modified (rough surfaces)
  - a. Non-reconstructive surgical approach
  - b. Reconstructive surgical approach (regardless the technique or materials used)
3. Mixed or unspecified surfaces
  - a. Non-reconstructive surgical approach
  - b. Reconstructive surgical approach (regardless the technique or materials used)

Findings were systematically summarized in tables according to pre-defined outcomes and subgroup analyses to enhance clarity and readability.

Within each treatment group, data were systematically collected on key parameters, including sample size (number of participants and implants), criteria used to define peri-implantitis, type of bone substitute, membrane used (if applicable), follow-up periods, implant system, and implant surface characteristics.

## 3 Results

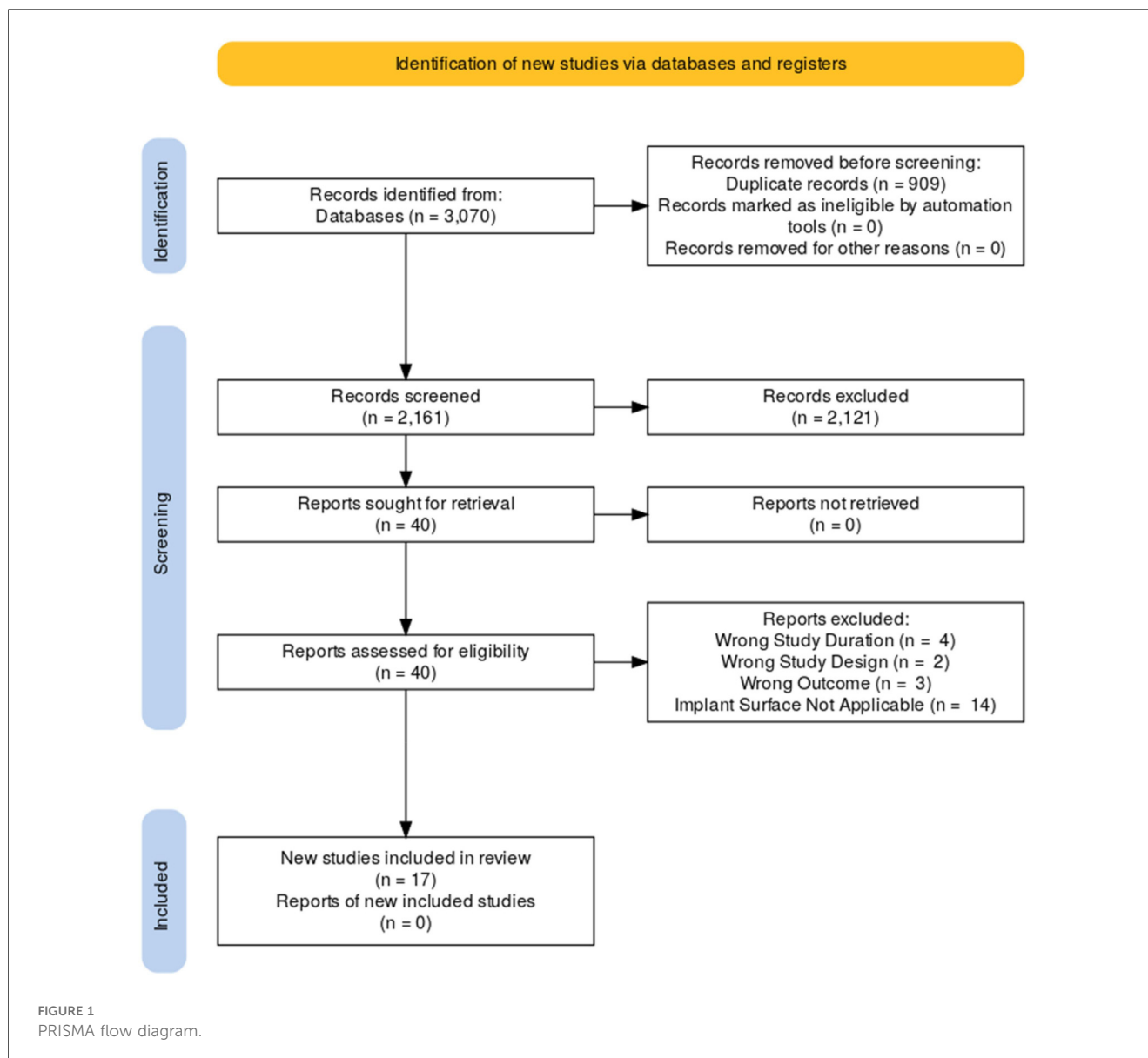
### 3.1 Study selection

The literature search process is illustrated in the flowchart below (Figure 1). In total, there are 17 studies included in the analysis (8–24). Among them, 8 are prospective cohort studies, 3 are retrospective cohort studies, 1 are randomized controlled trials. The remaining studies include 1 each of the following types: prospective clinical study, retrospective observational study, and prospective case series. A detailed description of the study characteristics can be found in the results in Tables 1a,b.

### 3.2 Study populations

#### 3.2.1 Peri-implantitis

Across the 17 studies analyzed, various diagnostic criteria have been employed to identify peri-implantitis, reflecting differences in study designs and clinical considerations (8–24). The most commonly reported diagnostic parameters include probing depth (PD), bleeding on probing (BOP), suppuration, and radiographic evidence of bone loss.



### 3.2.2 Probing depth (PD)

A probing depth threshold of  $\geq 6$  mm is frequently used as a criterion to identify peri-implantitis, as observed in studies by Carcuac et al., Romandini et al., and Rocuzzo et al. (15, 16, 26–28). Other studies, such as Aghazadeh et al. and Noelken et al., set a threshold of  $\geq 5$  mm, which is similar to the  $>4$  mm threshold considered indicative of disease by Mercado et al. and Schwarz et al. (24, 25, 31, 32). This variation highlights differences at diagnosis across studies.

### 3.2.3 Bleeding on probing (BOP) and suppuration

The presence of BOP and/or suppuration was consistently reported as a diagnostic marker in all studies. It serves as an indicator of ongoing inflammation and peri-implant tissue destruction. Studies such as La Monaca et al. and Khoury & Buchmann emphasize the importance of these clinical signs

in combination with radiographic findings for accurate diagnosis (20–22).

### 3.2.4 Radiographic bone loss

Radiographic evaluation of bone loss is another widely accepted criterion for peri-implantitis diagnosis. The threshold for bone loss varies among studies, with the most commonly reported value being  $\geq 3$  mm, as seen in Romandini et al. (16). Other studies, including Carcuac et al. and Aghazadeh et al., defined progressive bone loss based on post-treatment changes or specific defect characteristics, such as angular defects of  $\geq 3$  mm (15, 32). A more conservative threshold of  $\geq 1.8$  mm was applied in studies such as Roos-Jansåker et al., reflecting the variability in bone loss progression (29, 30).

TABLE 1a Summary of clinical studies evaluating treatments of peri-implantitis: surface types, materials, and outcomes.

| Author (year)                 | Implant surface                              | Implant brand                                                                     | Patients (initial/follow-up) | Implants (initial/follow-up)   | Type of treatment                                                                                         | Materials used                                                                                                                                           | Follow-up period | Surface impact                                                             |
|-------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------|------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------|
| Aghazadeh et al. (2022) (32)  | Mixed (Turned and medium-rough)              | Not explicitly stated, categorized by surface type                                | 45 (AB:22/16, BDX:23/23)     | 75 (AB:36/25, BDX:39/38)       | Reconstructive surgery (autogenous bone, xenograft, collagen membrane)                                    | Autogenous bone, xenograft (Bio-Oss), collagen membrane (OsseoGuard), titanium curettes, hydrogen peroxide, azithromycin, CHX rinse                      | 5 years          | No significant difference (turned vs. medium-rough surfaces)               |
| Carcuac et al. (2020) (15)    | Mixed (modified/non-modified)                | Not specified, classified as non-modified/modified                                | 100/73                       | 179/130                        | Open-flap debridement, surface decontamination, pocket elimination                                        | Systemic antibiotics, local antiseptics                                                                                                                  | 5 years          | Modified surfaces higher recurrence risk (OR 5.1, 95% CI: 1.6–16.5)        |
| Deppe et al. (2007) (18)      | Mixed (rough predominantly)                  | IMZ, Frialit-2, Bränemark, Straumann screw-type                                   | 32                           | 73 (Conventional:34, Laser:39) | Conventional vs. CO <sub>2</sub> laser-assisted (soft tissue resection/augmentation)                      | CO <sub>2</sub> laser, air-powder abrasive, $\beta$ -TCP/autogenous bone, Gore-Tex membrane                                                              | 5 years          | Superior outcomes with CO <sub>2</sub> laser in non-reconstructive therapy |
| Jemt & Eriksson (2021) (19)   | Mixed (Turned and moderately rough surfaces) | Bränemark (turned), TiUnite, Astra Tech OsseoSpeed, Lifecore RBM                  | 122 (initially 134, 12 lost) | 614/453                        | Non-reconstructive surgery (mechanical cleaning, osseous recontouring, antibiotics)                       | Mechanical debridement, hydrogen peroxide 10%, systemic antibiotics                                                                                      | Mean 7.3 years   | No significant difference (turned vs. moderately rough surfaces)           |
| Khoury & Buchmann (2001) (20) | Rough surfaces                               | IMZ, Frialit-2 (Friadent GmbH)                                                    | 25                           | 41                             | Reconstructive surgery (autogenous bone $\pm$ membranes)                                                  | Autogenous bone, ePTFE/Bioabsorbable barriers, CHX, citric acid, hydrogen peroxide, systemic antibiotics                                                 | 3 years          | No surface comparison (all rough)                                          |
| La Monaca et al. (2024) (22)  | Rough (TiUnite surface)                      | Nobel Biocare (Bränemark System, Göteborg)                                        | 34/23                        | 34/20                          | Reconstructive surgery (MDBA, resorbable membrane, chemical/mechanical decontamination)                   | MDBA (Puros), collagen membrane (Bio-Gide), hydrogen peroxide, CHX solution, tetracycline hydrochloride, systemic antibiotics (Amox/clav, Metronidazole) | 10 years         | All rough (TiUnite); no surface-specific comparison                        |
| La Monaca et al. (2018) (21)  | Rough (TiUnite surface)                      | Nobel Biocare (Bränemark System, Göteborg)                                        | 34                           | 34                             | Reconstructive surgical therapy (MDBA, resorbable collagen membrane, chemical/mechanical decontamination) | MDBA (Puros), resorbable membrane (Bio-Gide), hydrogen peroxide, CHX 0.2%, tetracycline hydrochloride, systemic antibiotics                              | 5 years          | All rough (TiUnite); no surface comparison                                 |
| Leonhardt et al. (2003) (23)  | Turned surfaces                              | Bränemark System (Nobel Biocare)                                                  | 9                            | 26                             | Surgical non-reconstructive therapy + systemic antibiotics                                                | Hydrogen peroxide 10%, individualized antibiotics (metronidazole, amoxicillin, tetracycline, ciprofloxacin, clindamycin), CHX rinse                      | 5 years          | All turned surfaces; not directly analyzed                                 |
| Mercado et al. (2018) (24)    | Rough (Micro-rough)                          | Bränemark TiUnite (46.66%), Astra Tech (26.66%), Straumann (10%), Others (16.66%) | 30                           | 30                             | Regenerative surgery (DBBMC, EMD, doxycycline, EDTA, ultrasonic scaler)                                   | DBBMC, EMD, doxycycline, EDTA 24%, ultrasonic scaler, chlorhexidine 0.12%                                                                                | 36 months        | Not specifically analyzed (micro-rough surfaces)                           |

(Continued)

TABLE 1a Continued

| Author (year)                    | Implant surface                            | Implant brand                                                                                  | Patients (initial/follow-up) | Implants (initial/follow-up)  | Type of treatment                                                                           | Materials used                                                                                                              | Follow-up period                  | Surface impact                                                        |
|----------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------|-------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------|
| Noelken et al. (2023) (25)       | • Mixed (rough predominantly)              | • Straumann, Ankylos, Brånemark, NobelActive, NobelPerfect, Frialit I, OsseoSpeed, Camlog, ICX | • 18                         | • 24                          | • LAPIDER (laser-assisted regeneration, autogenous bone, CT graft)                          | • Er:YAG laser, autogenous bone chips, doxycycline, CT graft, resorbable sutures                                            | • 3 years                         | • No direct comparison; rough surfaces only                           |
| Roccuzzo et al. (2017) (26)      | • Rough (SLA and TPS)                      | • Straumann (SLA, TPS)                                                                         | • 26/24                      | • 26/24                       | • Regenerative surgery (DBBMC, EDTA, CHX)                                                   | • DBBMC (Bio-Oss Collagen), EDTA 24%, CHX gel 1%, antibiotics (Amoxicillin/clavulanic acid)                                 | • 7 years                         | • SLA better clinical outcomes than TPS (significant)                 |
| Roccuzzo et al. (2020) (27)      | • Rough (SLA and TPS)                      | • Straumann (SLA, TPS)                                                                         | • 26/14                      | • 26/14                       | • Regenerative surgery (DBBMC, EDTA, CHX)                                                   | • DBBMC (Bio-Oss Collagen), EDTA 24%, CHX gel 1%, antibiotics (Amoxicillin/clavulanic acid)                                 | • 10 years                        | • SLA superior survival/ outcomes compared to TPS                     |
| Roccuzzo et al. (2021) (28)      | • Rough (SLA)                              | • Straumann (SLA implants)                                                                     | • 75/51                      | • 75/64                       | • Reconstructive surgery (DBBMC, EDTA, CHX gel, titanium curettes, connective tissue graft) | • DBBMC, EDTA 24%, CHX 1% gel, titanium curettes, titanium brush, connective tissue graft, systemic antibiotics             | • 5 years                         | • Uniform (all SLA surfaces)                                          |
| Romandini et al. (2024) (16)     | • Mixed (Turned and Modified)              | • Nobel Biocare (74.2%), Astra Tech (18%), Straumann (6.7%), Neoss (1.1%)                      | • 149                        | • 267                         | • Non-reconstructive surgical therapy (titanium-coated curettes, systemic antibiotics)      | • Titanium-coated curettes, saline/CHX gauze, selective systemic antibiotics (Amoxicillin)                                  | • Mean 7 years (range 1-18 years) | • Modified surfaces significantly higher implant loss risk (HR = 4.5) |
| Roos-Jansäker et al. (2011) (29) | • Primarily machined, few modified (rough) | • Brånemark (majority), Astra Tech (minority)                                                  | • 38/32                      | • 65/56                       | • Reconstructive surgery (Aligipore ± membrane)                                             | • Aligipore, Ossequest membrane, H <sub>2</sub> O <sub>2</sub> 3%, antibiotics (amoxicillin, metronidazole), CHX 0.1% rinse | • 3 years                         | • No significant impact reported (mostly machined)                    |
| Roos-Jansäker et al. (2014) (30) | • Primarily machined, few modified (rough) | • Brånemark (majority), Astra Tech (minority)                                                  | • 38/25                      | • Not initially specified/ 45 | • Reconstructive surgery (Aligipore ± membrane)                                             | • Aligipore, Ossequest membrane, H <sub>2</sub> O <sub>2</sub> 3%, antibiotics (amoxicillin, metronidazole), CHX rinse      | • 5 years                         | • No significant impact (mostly machined surfaces)                    |
| Schwarz et al. (2009) (31)       | • Mixed (machined and rough surfaces)      | • Brånemark, Camlog, ITI, KSI Bauer Schraube, Zimmer, ZL-Duraplast                             | • 22/19                      | • 22/19                       | • Regenerative surgery (NBM + CM vs. NHA)                                                   | • NBM (BioOss), CM (BioGide), NHA (Ostim), plastic curettes, CHX 0.2%                                                       | • 48 months                       | • Surface-specific outcomes not detailed                              |

TABLE 1b Comparative outcomes of clinical studies on peri-implantitis treatments: implant surfaces, secondary outcomes, and long-term surface impact.

| Author (year)                 | Implant loss (%)                                                                                   | Disease resolution                                                                                                                 | Radiographic bone loss/gain                                                                                     | Mean PD post-treatment                                                                                            | Subgroup analysis (by surface & approach)                                                                                                          | Conclusions surface impact                                                                                                                                                                                                   |
|-------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aghazadeh et al. (2022) (32)  | <ul style="list-style-type: none"><li>AB: 1 fractured, BDX: 1 fractured (both initially)</li></ul> | <ul style="list-style-type: none"><li>PD reduction AB: 1.7 mm, BDX: 2.8 mm at 5 years</li></ul>                                    | <ul style="list-style-type: none"><li>AB: -0.7 mm (loss), BDX: + 1.6 mm (gain)</li></ul>                        | <ul style="list-style-type: none"><li>AB: PD reduced by 1.7 mm, BDX: by 2.8 mm at 5 years</li></ul>               | <ul style="list-style-type: none"><li>BDX superior to AB; reconstructive approach only, no surface-specific outcomes detailed</li></ul>            | <ul style="list-style-type: none"><li>BDX superior to AB; supportive maintenance every 3 months emphasized.No significant difference (turned vs. medium-rough surfaces)</li></ul>                                            |
| Carcuac et al. (2020) (15)    | <ul style="list-style-type: none"><li>20.8%</li></ul>                                              | <ul style="list-style-type: none"><li>Not explicitly combined; PD ≥ 6 mm at 1 year = increased recurrence (OR 7.4)</li></ul>       | <ul style="list-style-type: none"><li>Stable from year 1; further loss &gt; 1 mm in 13.1%</li></ul>             | <ul style="list-style-type: none"><li>5.0 mm (stable from year 1 at 4.9 mm)</li></ul>                             | <ul style="list-style-type: none"><li>Non-modified: 17% recurrence, Modified: 52% recurrence; no explicit surgical subgroup</li></ul>              | <ul style="list-style-type: none"><li>Surgical treatment effective, but 44% recurrence; modified surfaces, deep residual PD major risk factors.Modified surfaces higher recurrence risk (OR 5.1, 95% CI: 1.6–16.5)</li></ul> |
| Deppe et al. (2007) (18)      | <ul style="list-style-type: none"><li>17.8%</li></ul>                                              | <ul style="list-style-type: none"><li>Superior PD reduction laser (1.0–2.4 mm at 4 months)</li></ul>                               | <ul style="list-style-type: none"><li>Laser better short-term bone gain; long-term similar outcomes</li></ul>   | <ul style="list-style-type: none"><li>Laser: 1.0–2.4 mm, Conventional: 2.4–7.9 mm (long-term)</li></ul>           | <ul style="list-style-type: none"><li>Laser superior in soft tissue resection; similar in augmentation</li></ul>                                   | <ul style="list-style-type: none"><li>Laser beneficial in non-reconstructive procedures; similar long-term results in augmentation.Superior outcomes with CO<sub>2</sub> laser in non-reconstructive therapy</li></ul>       |
| Jemt & Eriksson (2021) (19)   | <ul style="list-style-type: none"><li>16.7%</li></ul>                                              | <ul style="list-style-type: none"><li>Not explicitly reported; primarily bone-level outcomes</li></ul>                             | <ul style="list-style-type: none"><li>Increased bone loss post-treatment (0.26 mm/year)</li></ul>               | <ul style="list-style-type: none"><li>PD not explicitly reported</li></ul>                                        | <ul style="list-style-type: none"><li>No significant differences between surface types; increased bone loss post-treatment</li></ul>               | <ul style="list-style-type: none"><li>Surgical treatment ineffective long-term; bone loss accelerated post-treatment, edentulous worse prognosis.No significant difference (turned vs. moderately rough surfaces)</li></ul>  |
| Khoury & Buchmann (2001) (20) | <ul style="list-style-type: none"><li>0%</li></ul>                                                 | <ul style="list-style-type: none"><li>PD improvement: Bone alone 8.0→2.9 mm, worse with membranes (7.7→5.1 mm)</li></ul>           | <ul style="list-style-type: none"><li>Significant bone gain (bone alone 3.2 mm, membranes lower)</li></ul>      | <ul style="list-style-type: none"><li>Bone alone: 2.9 mm; Non-resorbable: 2.8 mm; Bioabsorbable: 5.1 mm</li></ul> | <ul style="list-style-type: none"><li>Bone alone superior; membranes caused high complications</li></ul>                                           | <ul style="list-style-type: none"><li>Bone grafting alone effective; membranes did not enhance outcomes, increased complications.No surface comparison (all rough)</li></ul>                                                 |
| La Monaca et al. (2024) (22)  | <ul style="list-style-type: none"><li>8.8%</li></ul>                                               | <ul style="list-style-type: none"><li>Composite success: 53%; PD significantly reduced to 2.95 mm at 10 years</li></ul>            | <ul style="list-style-type: none"><li>Mean bone gain: 1.07 mm (stable 10 years)</li></ul>                       | <ul style="list-style-type: none"><li>Reduced from 6.33 mm to 2.95 mm at 10 years</li></ul>                       | <ul style="list-style-type: none"><li>Reconstructive approach only; stable bone gain, PD improvement, composite success 53%</li></ul>              | <ul style="list-style-type: none"><li>Long-term reconstructive therapy stable; supportive therapy crucial, defect severity not significant.All rough (TiUnit); no surface-specific comparison</li></ul>                      |
| La Monaca et al. (2018) (21)  | <ul style="list-style-type: none"><li>0%</li></ul>                                                 | <ul style="list-style-type: none"><li>PD reduced initially, increased by 5 years (4.62 mm, nonsignificant from baseline)</li></ul> | <ul style="list-style-type: none"><li>Initial significant bone gain (1 year), gradual loss by 5 years</li></ul> | <ul style="list-style-type: none"><li>4.62 mm at 5 years (from 5.93 mm baseline)</li></ul>                        | <ul style="list-style-type: none"><li>Reconstructive surgery initially effective, unstable outcomes long-term</li></ul>                            | <ul style="list-style-type: none"><li>Initial benefits lost over time, no implants lost; unpredictable outcomes long-term.All rough (TiUnit); no surface comparison</li></ul>                                                |
| Leonhardt et al. (2003) (23)  | <ul style="list-style-type: none"><li>27%</li></ul>                                                | <ul style="list-style-type: none"><li>Significant reduction plaque (100%→11%) and bleeding (100%→5%)</li></ul>                     | <ul style="list-style-type: none"><li>Stable in 9 implants, gain in 6 implants, loss in 4 implants</li></ul>    | <ul style="list-style-type: none"><li>Not explicitly reported; clinical signs significantly improved</li></ul>    | <ul style="list-style-type: none"><li>Non-reconstructive approach; success 58%, significant clinical improvements</li></ul>                        | <ul style="list-style-type: none"><li>Limited success (58%); smoking negatively impacted outcomes.All rough surfaces; not directly analyzed</li></ul>                                                                        |
| Mercado et al. (2018) (24)    | <ul style="list-style-type: none"><li>0%</li></ul>                                                 | <ul style="list-style-type: none"><li>Mean PD from 8.9 mm to 3.5 mm; BOP/suppression from 100% to 20%</li></ul>                    | <ul style="list-style-type: none"><li>Bone gain (6.92 mm to 2.60 mm mean bone loss at 36 months)</li></ul>      | <ul style="list-style-type: none"><li>Reduced from 8.9 mm to 3.5 mm at 36 months</li></ul>                        | <ul style="list-style-type: none"><li>Regenerative (reconstructive) approach only, significant PD reduction and bone gain, success 56.7%</li></ul> | <ul style="list-style-type: none"><li>Regenerative treatment effective (56.7% success); regular supportive therapy critical.Not specifically analyzed (micro-rough surfaces)</li></ul>                                       |

(Continued)

TABLE 1b Continued

| Author (year)                    | Implant loss (%)                                                                        | Disease resolution                                                                                                              | Radiographic bone loss/gain                                                                                                         | Mean PD post-treatment                                                                                             | Subgroup analysis (by surface & approach)                                                                                                              | Conclusions surface impact                                                                                                                                                                                                    |
|----------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Noelken et al. (2023) (25)       | <ul style="list-style-type: none"><li>8.3%</li></ul>                                    | <ul style="list-style-type: none"><li>PD significantly reduced (5.05 to 3.08 mm), BOP 100% to 36.4%</li></ul>                   | <ul style="list-style-type: none"><li>Significant bone gain (Interproximal 3.1 mm, Buccal 3.5 mm, Lingual 1.46 mm)</li></ul>        | <ul style="list-style-type: none"><li>Final PD 3.08 mm</li></ul>                                                   | <ul style="list-style-type: none"><li>LAPIDER effective for severe defects; significant hard/soft tissue regeneration</li></ul>                        | <ul style="list-style-type: none"><li>LAPIDER provided substantial regeneration and aesthetic improvements.No direct comparison; rough surfaces only</li></ul>                                                                |
| Roccuzzo et al. (2017) (26)      | <ul style="list-style-type: none"><li>16.7% overall (SLA:16.7%, TPS:28.6%)</li></ul>    | <ul style="list-style-type: none"><li>Significant PD reduction, SLA (3.2 mm), TPS (3.4 mm); BOP SLA (7.5%), TPS (30%)</li></ul> | <ul style="list-style-type: none"><li>Significant bone fill SLA (2.1 mm gain), TPS (2.0 mm gain)</li></ul>                          | <ul style="list-style-type: none"><li>SLA: 3.2 mm, TPS: 3.4 mm</li></ul>                                           | <ul style="list-style-type: none"><li>Regenerative approach better for SLA vs. TPS implants</li></ul>                                                  | <ul style="list-style-type: none"><li>DBBMC effective; SLA surfaces significantly better clinical outcomes than TPS.</li></ul>                                                                                                |
| Roccuzzo et al. (2020) (27)      | <ul style="list-style-type: none"><li>SLA:20%, TPS:45%</li></ul>                        | <ul style="list-style-type: none"><li>Significant PD reduction SLA (3.2 mm), TPS (3.4 mm); BOP significantly reduced</li></ul>  | <ul style="list-style-type: none"><li>SLA substantial gain (2.7 mm), TPS moderate gain (2.0 mm)</li></ul>                           | <ul style="list-style-type: none"><li>SLA: 3.2 mm, TPS: 3.5 mm</li></ul>                                           | <ul style="list-style-type: none"><li>SLA implants superior long-term survival/outcomes</li></ul>                                                      | <ul style="list-style-type: none"><li>DBBMC stable outcomes; SLA significantly better than TPS long-term</li></ul>                                                                                                            |
| Roccuzzo et al. (2021) (28)      | <ul style="list-style-type: none"><li>17%</li></ul>                                     | <ul style="list-style-type: none"><li>PD 6.89 mm to 4.06 mm; BOP 70.6% to 17.2%</li></ul>                                       | <ul style="list-style-type: none"><li>Substantial bone fill; no numeric specifics reported</li></ul>                                | <ul style="list-style-type: none"><li>Reduced from 6.89 mm to 4.06 mm at 5 years</li></ul>                         | <ul style="list-style-type: none"><li>Reconstructive approach only; SLA implants only, survival 80%, success 45.3%, significant PD reduction</li></ul> | <ul style="list-style-type: none"><li>Reconstructive protocol effective; adherence to supportive therapy significantly improves outcomes.</li></ul>                                                                           |
| Romandini et al. (2024) (16)     | <ul style="list-style-type: none"><li>19.9%</li></ul>                                   | <ul style="list-style-type: none"><li>Not explicitly; high recurrence, retreatment common (24.3%)</li></ul>                     | <ul style="list-style-type: none"><li>Mean additional bone loss: 0.97 mm; &gt;1 mm loss in 42.4%</li></ul>                          | <ul style="list-style-type: none"><li>Not explicitly detailed post-treatment; baseline deepest PD 7.8 mm</li></ul> | <ul style="list-style-type: none"><li>Turned surfaces significantly better prognosis; modified surfaces high loss risk (HR = 4.5)</li></ul>            | <ul style="list-style-type: none"><li>High recurrence; surface type crucial predictor, severe baseline bone loss/suppurative increase loss risk.Modified surfaces significantly higher implant loss risk (HR = 4.5)</li></ul> |
| Roos-Jansäker et al. (2011) (29) | <ul style="list-style-type: none"><li>0%</li></ul>                                      | <ul style="list-style-type: none"><li>Clinical measures not explicitly detailed</li></ul>                                       | <ul style="list-style-type: none"><li>Stable bone fill: bone graft (1.3 mm), membrane (1.6 mm), no significant difference</li></ul> | <ul style="list-style-type: none"><li>Not explicitly reported</li></ul>                                            | <ul style="list-style-type: none"><li>Stable outcomes, no significant membrane advantage</li></ul>                                                     | <ul style="list-style-type: none"><li>Aligipore ± membrane effective, stable bone fill, membrane no additional advantage.No significant impact reported (mostly machined)</li></ul>                                           |
| Roos-Jansäker et al. (2014) (30) | <ul style="list-style-type: none"><li>0%</li></ul>                                      | <ul style="list-style-type: none"><li>PD reduction: 3.0–3.3 mm, BOP significantly reduced</li></ul>                             | <ul style="list-style-type: none"><li>Bone gain stable: 1.1–1.3 mm, no significant membrane advantage</li></ul>                     | <ul style="list-style-type: none"><li>2.6–2.7 mm</li></ul>                                                         | <ul style="list-style-type: none"><li>No significant advantage with membrane</li></ul>                                                                 | <ul style="list-style-type: none"><li>Stable results; membrane no additional advantage over bone substitute alone.No significant impact (mostly machined surfaces).</li></ul>                                                 |
| Schwarz et al. (2009) (31)       | <ul style="list-style-type: none"><li>0% explicitly reported (1 discontinued)</li></ul> | <ul style="list-style-type: none"><li>NBM + CM: PD reduction 2.5 mm, NHA: 1.1 mm, BOP significantly reduced</li></ul>           | <ul style="list-style-type: none"><li>NBM + CM superior bone fill compared to NHA</li></ul>                                         | <ul style="list-style-type: none"><li>NBM + CM: 4.6 mm, NHA: 5.8 mm</li></ul>                                      | <ul style="list-style-type: none"><li>NBM + CM significantly better than NHA</li></ul>                                                                 | <ul style="list-style-type: none"><li>NBM + CM superior long-term clinical/radiographic outcomes vs. NHA.Surface-specific outcomes not detailed.</li></ul>                                                                    |

AB, autogenous bone; BDX, bovine-derived xenograft; BOP, bleeding on probing; CAL, clinical attachment level; CHX, chlorhexidine; CI, confidence interval; CM, collagen membrane; CO<sub>2</sub>, carbon dioxide; CT, connective tissue; DBBMC, deproteinized bovine bone mineral with collagen; DH, defect height; EMD, enamel matrix derivative; ePTFE, expanded polytetrafluoroethylene; H<sub>2</sub>O<sub>2</sub>, hydrogen peroxide; HR, hazard ratio; LAPIDER, laser-assisted peri-implant defect regeneration; MBL, marginal bone level; MDBA, mineralized dehydrated bone allograft; NBM, natural bone mineral; NHA, nanocrystalline hydroxyapatite; OR, odds ratio; PD, probing depth; P-IS, peri-implantitis surgery; SLA, sandblasted large grit acid-etched; SOP, suppurative on probing; TPS, titanium plasma-sprayed; β-TCP, beta-tricalcium phosphate.

### 3.2.5 Variability in diagnostic criteria

Despite a general agreement on the primary diagnostic signs—probing depth, BOP/suppuration, and radiographic bone loss—variability exists in the specific thresholds and additional criteria applied across studies.

A detailed overview of the case definitions used to include patients with peri-implantitis in each study (treatment group) is provided in [Table 2](#).

### 3.3 Primary outcome: recurrence and treatment failure

The included studies demonstrated that implant surface characteristics influenced recurrence rates following surgical peri-implantitis treatment. Modified (rough) surfaces consistently showed higher recurrence compared with turned (machined) surfaces. Carcuac et al. reported an overall recurrence of 44%, with a significantly increased risk for modified surfaces (OR 5.1) (15). Similarly, Romandini et al. found a retreatment rate of 24.3%. In contrast, studies involving turned surfaces, such as Leonhardt et al., reported more stable outcomes (23). These findings indicate that surface roughness is a key determinant of recurrence and long-term treatment stability. Studies by Schwarz et al., Mercado et al. and Noelken

et al. documented relatively stable outcomes without explicitly reporting significant recurrence rates (24, 25, 31).

### 3.4 Secondary outcomes

#### 3.4.1 Implant loss

Implant loss was more frequent among rough surface implants, especially TPS, with Rocuzzo et al. reporting loss in 45% of TPS implants (27), and Leonhardt et al. reporting 27% for turned surfaces (23). SLA surfaces demonstrated better survival than TPS, with 20% vs. 45% loss after 10 years (27). Modified surfaces were identified as a strong predictor of implant loss (HR 4.5) (16). Turned surfaces generally exhibited lower long-term loss risk, around 20%, compared to modified ones (16). Lower implant loss rates were generally associated with reconstructive surgical approaches, as observed by Noelken et al. 8.3% (25) and La Monaca et al. (22), 8.8%.

#### 3.4.2 Disease resolution and probing depth (PD)

Reconstructive surgery generally improved PD irrespective of surface, but rough surfaces demonstrated greater variability. Mercado et al. (2018) reported PD reduction from 8.9 mm to 3.5 mm on micro-rough implants (24), while Noelken et al. achieved PD reduction from 5.05 mm to 3.08 mm in

TABLE 2 Definitions of periimplantitis.

| Study                           | Definition of periimplantitis                                                                                                                                                                                                      |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aghazadeh et al., 2022 (32)     | Probing pocket depths of at least 5 mm, presence of bleeding on probing and/or suppuration, radiographic bone loss of 2 mm or more from implant placement to screening, and an angular peri-implant bone defect of 3 mm or greater |
| Carcuac et al., 2020 (15)       | Probing pocket depths of 6 mm or more, presence of bleeding on probing, reduced marginal bone level and progressive bone loss greater than 1 mm post-treatment                                                                     |
| Deppe et al., 2007 (18)         | Probing pocket depths of at least 5 mm, presence of bleeding on probing, radiographic evidence of progressive vertical bone loss, and clinical signs of inflammation                                                               |
| Jemt et al., 2021 (19)          | Bone loss exceeding 0.4 mm, mucosal inflammation, presence of plaque and/or suppuration, and radiographic evidence of marginal bone loss                                                                                           |
| Khoury et al., 2001 (20)        | Bone loss of more than 50% of the implant length, augmented probing depths, bleeding on probing, and radiographic evidence of intrabony defects                                                                                    |
| La Monaca et al., 2018 (21)     | Progressive bone loss of 3 mm or more detected on radiographs, the presence of bleeding on probing and/or suppuration, and probing depths of at least 5 mm                                                                         |
| La Monaca et al., 2024 (22)     | Progressive angular bone loss of at least 3 mm beyond crestal bone level changes, the presence of bleeding on gentle probing and/or suppuration, and implants in function for more than 12 months                                  |
| Leonhardt et al., 2003 (23)     | Marginal bone loss of at least three implant threads compared to baseline radiographs, bleeding on probing and/or suppuration from peri-implant sulci, and microbiological confirmation of peri-implant pathogens                  |
| Mercado et al., 2018 (24)       | Probing pocket depths exceeding 4 mm, the presence of bleeding on probing and/or suppuration, a minimum radiographic bone loss of 20%, and implants that have been in function for at least 2 years                                |
| Noelken et al., 2023 (25)       | Probing pocket depths greater than 5 mm, the presence of bleeding on probing and suppuration, and radiographically confirmed bone loss                                                                                             |
| Rocuzzo et al., 2017 (26)       | Probing pocket depths of at least 6 mm, no implant mobility, bleeding on probing and/or suppuration, and radiographic bone loss exceeding three implant threads compared to baseline                                               |
| Rocuzzo et al., 2020 (27)       | Probing pocket depths of 6 mm or greater, the presence of bleeding on probing and/or suppuration, radiographic bone loss beyond crestal changes, and the absence of implant mobility                                               |
| Rocuzzo et al., 2021 (28)       | Probing pocket depths reach or exceed 6 mm, bleeding on probing, radiographic evidence of progressive bone loss, and the presence of pus or inflammation                                                                           |
| Romandini et al., 2024 (16)     | Probing pocket depths of 6 mm or more, bleeding and/or suppuration on probing, and radiographic evidence of marginal bone loss equal to or greater than 3 mm                                                                       |
| Roos-Jansäker et al., 2011 (29) | Radiographic bone loss of at least 1.8 mm following the first year in function, the presence of bleeding and/or pus on probing, and inclusion criteria of non-mobile implants                                                      |
| Roos-Jansäker et al., 2014 (30) | Radiographic bone loss of at least 3 threads (≥1.8 mm), the presence of a vertical defect component, and bleeding on probing and/or suppuration                                                                                    |
| Schwarz et al., 2009 (31)       | Probing pocket depths greater than 4 mm, presence of bleeding on probing and/or suppuration, radiographic evidence of bone loss, and an intrabony defect component of at least 3 mm                                                |

TABLE 3a Risk of bias assessment for RCTs: RoB 2 risk of bias assessment.

| Study             | Bias arising from the randomization process | Bias due to deviations from intended interventions | Bias due to missing outcome data | Bias in measurement of the outcome | Bias in selection of the reported result | Overall bias  |
|-------------------|---------------------------------------------|----------------------------------------------------|----------------------------------|------------------------------------|------------------------------------------|---------------|
| Schwarz 2009 (31) | Low                                         | Low                                                | Some concerns                    | Low                                | Low                                      | Some concerns |

predominantly rough implants (25) identifying disease resolution. Rocuzzo et al. observed significant PD improvements for SLA implants compared with TPS (26, 27). Turned implants Leonhardt et al., also demonstrated significant PD reduction (23). Conversely, non-reconstructive surgical approaches such as that of Deppe et al. with predominantly rough surfaces showed initial short-term PD reductions with inconsistent long-term stability (18).

### 3.4.3 Radiographic bone changes

Bone regeneration outcomes were surface-dependent. Reconstructive procedures around rough implants, particularly SLA, showed consistent bone gain (Rocuzzo et al. + 2.1 mm; + 2.7 mm) (26, 27). TPS implants demonstrated less favorable long-term stability, even with grafting (27). Smooth (turned) surfaces were rarely evaluated in regenerative contexts, limiting conclusions. Khoury & Buchmann reported substantial bone gain (3.2 mm) on rough implants with autografts (20), while Roos-Jansäker et al. found stable bone gain (1.1–1.6 mm) in predominantly machined implants (29, 30).

## 3.5 Subgroup analyses

### 3.5.1 Turned surfaces

Showed moderate long-term stability, but implant loss remained high when treated non-reconstructively (23). Reconstructive data were limited but suggested stable outcomes (29, 30).

### 3.5.2 Modified (rough) surfaces

Non-reconstructive approaches resulted in high recurrence and implant loss (15, 16). Reconstructive approaches improved outcomes, with SLA surfaces outperforming TPS (26, 27).

### 3.5.3 Mixed surfaces

Outcomes were heterogeneous. Laser-assisted non-reconstructive therapy demonstrated short-term benefits Deppe et al. (18), but Jemt & Eriksson reported long-term bone loss regardless of surface type (19). Reconstructive treatments showed better results with natural bone mineral combined with a collagen membrane (NBM + CM) compared to nanocrystalline hydroxyapatite (NHA) (31), but surface-specific differences remained underreported. Aghazadeh et al. reported improved outcomes with xenograft (BDX) usage (32).

The detailed study characteristics and outcomes are presented in Tables 1a,b.

## 3.6 Risk of bias

The risk of bias assessment for the 17 included studies highlights several concerns across different domains (Tables 3a and b).

The RCT (Schwarz 2009) had some concerns (31).

Among the non-randomized studies, serious risk of bias was frequently noted in selection of participants e.g., Aghazadeh, Carcuac, and La Monaca (15, 17, 22) and missing data e.g., Rocuzzo, Romandini, Roos-Jansaker (16, 27, 30). However, intervention classification and reporting of results were generally at low risk across studies. Confounding and deviations from interventions were rated as moderate risk in most cases.

Ten studies were judged to have a serious overall bias, primarily due to participant selection and missing data. The remaining studies had a moderate overall bias, with issues mainly related to confounding and missing data.

Certainty of evidence for the main outcomes was assessed using the GRADE approach and is presented in Supplementary Table S1. Overall, the certainty was judged to be very low to low across all outcomes, primarily due to serious risk of bias, high heterogeneity of diagnostic criteria and outcome definitions, small sample sizes, and imprecision of effect estimates.

## 4 Discussion

This systematic review aimed to evaluate the impact of implant surface characteristics on the long-term outcomes of surgical treatment of peri-implantitis. The main findings indicate that modified (rough) surfaces are consistently associated with higher recurrence and implant loss compared with turned (machined) surfaces. Within rough surfaces, SLA implants achieved more favorable outcomes than TPS, particularly in reconstructive contexts (15, 16, 26–28). In contrast, smooth implants demonstrated comparatively lower recurrence (23). These results underscore that implant surface topography can be a determinant of surgical treatment prognosis.

The effectiveness of peri-implantitis surgery is influenced by both treatment modality and implant surface. For non-regenerative procedures, modified surfaces were repeatedly associated with worse outcomes: Carcuac et al. reported a 44% recurrence rate with rough implants (15), while Romandini et al. identified modified surfaces such as TiUnite and SLA as predictors of implant loss (16). In contrast, turned surfaces showed more stable disease suppression (23).

For regenerative approaches, implant surface also played a key role. SLA implants demonstrated favorable long-term bone gain

TABLE 3b Risk of bias assessment for non- RCTs: ROBINS-i risk of bias assessment.

| Study                   | Bias due to confounding | Bias in selection of participants into the study | Bias in classification of interventions | Bias due to deviations from intended interventions | Bias due to missing data | Bias in measurement of outcomes | Bias in selection of reported results | Overall bias |
|-------------------------|-------------------------|--------------------------------------------------|-----------------------------------------|----------------------------------------------------|--------------------------|---------------------------------|---------------------------------------|--------------|
| Aghazadeh 2022 (32)     | Moderate                | Serious                                          | Low                                     | Moderate                                           | Serious                  | Moderate                        | Low                                   | Serious      |
| Carcuac 2020 (15)       | Moderate                | Serious                                          | Low                                     | Moderate                                           | High                     | Low                             | Low                                   | Serious      |
| Deppe 2007 (18)         | Moderate                | Moderate                                         | Low                                     | Moderate                                           | Serious                  | Moderate                        | Low                                   | Serious      |
| Jemt et al. 2021 (19)   | Moderate                | Serious                                          | Low                                     | Moderate                                           | Serious                  | Moderate                        | Low                                   | Serious      |
| Khoury 2001 (20)        | Serious                 | Low                                              | Moderate                                | Moderate                                           | Low                      | Moderate                        | Low                                   | Moderate     |
| La Monaca 2018 (21)     | Moderate                | Moderate                                         | Low                                     | Moderate                                           | Serious                  | Low                             | Low                                   | Moderate     |
| La Monaca 2024 (22)     | Serious                 | Serious                                          | Low                                     | Moderate                                           | Serious                  | Moderate                        | Low                                   | Serious      |
| Leonhardt 2003 (23)     | Moderate                | Low                                              | Low                                     | Moderate                                           | Low                      | Moderate                        | Low                                   | Moderate     |
| Mercado 2018 (24)       | Moderate                | Low                                              | Low                                     | Moderate                                           | Low                      | Low                             | Low                                   | Moderate     |
| Noelken 2023 (25)       | Moderate                | Moderate                                         | Low                                     | Moderate                                           | Serious                  | Moderate                        | Low                                   | Moderate     |
| Roccuzzo 2017 (26)      | Moderate                | Moderate                                         | Low                                     | Moderate                                           | Moderate                 | Low                             | Low                                   | Moderate     |
| Roccuzzo 2020 (27)      | Moderate                | Serious                                          | Low                                     | Moderate                                           | Serious                  | Low                             | Low                                   | Serious      |
| Roccuzzo 2021 (28)      | Moderate                | Serious                                          | Low                                     | Moderate                                           | Serious                  | Moderate                        | Low                                   | Serious      |
| Romandini 2024 (16)     | Moderate                | Serious                                          | Low                                     | Moderate                                           | Serious                  | Moderate                        | Low                                   | Serious      |
| Roos-Jansaker 2011 (29) | Moderate                | Serious                                          | Low                                     | Moderate                                           | Serious                  | Low                             | Low                                   | Serious      |
| Roos-Jansaker 2014 (30) | Moderate                | Serious                                          | Low                                     | Moderate                                           | Serious                  | Low                             | Low                                   | Serious      |

and PD reduction (26–28), whereas TPS implants performed poorly even when grafting was applied (27). Although smooth implants were less frequently studied in regenerative contexts, available evidence suggests they may perform adequately when combined with supportive therapy (29, 30).

Bone regeneration outcomes differed substantially according to surface characteristics. Greater bone fill was generally reported around rough surfaces when grafting materials were used. Khoury and Buchmann observed a 2.4 mm gain at 12 months using autogenous grafts on rough implants (20). Roccuzzo et al. reported significant defect reduction with xenografts, particularly in SLA implants, while TPS implants showed limited stability (26, 27). Comparable results with alloplastic materials were also noted (33, 34). However, bone regeneration around smooth surfaces was less favorable: Roos-Jansåker et al. reported limited improvement with alloplastic grafts (30). Thus, while rough implants may predispose to recurrence, they also appear to support more pronounced bone regeneration after reconstructive procedures.

This paradox may be explained by surface-related biology. Rough surfaces are harder to decontaminate and accumulate more plaque (35, 36), yet they may stabilize the coagulum and promote defect fill (37). Accordingly, radiographic bone gain does not necessarily correspond to re-osseointegration, as several animal studies identified connective tissue interposition rather than true reattachment (38–40).

The role of membranes in guided bone regeneration (GBR) has also been linked to implant surfaces. Khoury et al. showed greater bone gain with non-resorbable membranes around rough implants (20), while Deppe et al. observed comparable results with resorbable membranes (18). These data suggest that both membrane type and surface roughness influence regenerative outcomes. Furthermore, clinical studies and experimental models in dogs indicate that rough surfaces generally achieve greater defect fill than smooth surfaces under GBR conditions (37).

Surface characteristics may also impact soft tissue attachment. Excessively smooth surfaces can impair mucosal adhesion, as Quirynen et al. observed attachment loss on polished abutments compared with stable CAL around commercially available surfaces (41). Other studies support that maintaining a certain degree of roughness enhances soft tissue sealing (42). These findings provide a biological explanation for the improved clinical outcomes of rough implants after GBR, despite their higher susceptibility to recurrence.

Interpretation of the evidence is complicated by considerable heterogeneity. Defect morphology influences outcomes, with narrower defects showing better results (17, 43), yet most studies failed to provide detailed descriptions, limiting cross-study comparisons. Moreover, peri-implantitis definitions varied widely: Roccuzzo et al. required  $\geq 6$  mm PD and bone loss exceeding three implant threads (26), while Mercado et al. used  $\geq 4$  mm PD and  $\geq 20\%$  radiographic bone loss (24). Measurement variability further complicates interpretation (44). Such inconsistencies directly affect assessment of surface-related outcomes and hinder robust comparisons across studies.

This review is limited by the substantial heterogeneity among the included studies, particularly in peri-implantitis diagnostic

criteria, defect morphology, surgical techniques, and outcome measures. Most studies were small in size, lacked standardized definitions, and many were judged to have a serious overall risk of bias, especially in participant selection and missing data. Confounding variables were insufficiently controlled, further reducing certainty. Furthermore, the restriction to English-language studies may have introduced language bias, potentially leading to omission of relevant non-English publications. Applying the GRADE framework, the certainty of the available evidence was rated as very low to low for all main outcomes, reflecting methodological shortcomings and heterogeneity among the included studies. These limitations restrict the generalizability of the findings and reinforce the need for well-designed, adequately powered randomized controlled trials with standardized definitions and longer follow-up.

## 5 Conclusion

The effectiveness of peri-implantitis surgery is influenced by implant surface characteristics and treatment modality. Modified surfaces are generally more prone to recurrence and implant loss, with SLA implants performing better than TPS, while turned surfaces appear less susceptible but remain insufficiently studied in regenerative contexts. Reconstructive approaches combined with supportive care consistently provide the most favorable outcomes. Given the very low to low certainty of the evidence with heterogeneous results, current findings should be interpreted with caution, and well-designed long-term randomized trials with standardized definitions and consistent surface classifications are urgently needed. Future trials should adopt standardized outcome definitions (e.g., PD thresholds, BOP, radiographic bone loss criteria) to allow comparability across studies. Research should focus on RCTs directly comparing surface types, long-term follow-up, and adjustment for confounding factors such as defect morphology and maintenance compliance. Addressing these gaps will clarify the role of implant surface modifications.

### 5.1 Clinical implications

When planning peri-implantitis surgery, implant surface characteristics should be taken into account, but they must not be considered in isolation. Evidence indicates that reconstructive approaches yield more reliable outcomes than non-reconstructive ones, particularly for rough implants, with SLA surfaces performing more favorably than TPS. Turned (machined) surfaces appear less prone to recurrence, although data on regenerative protocols remain scarce. These observations suggest that implant surface may influence prognosis, yet it represents only one part of a complex clinical picture.

Patient-related risk factors (such as smoking, systemic conditions, low compliance and/or adherence to supportive care) exert a profound effect on long-term success and may outweigh surface-related differences. Surgical decision-making should therefore be individualized, integrating implant surface

type, defect morphology, patient risk profile, and anticipated compliance. The use of biomaterials and barrier membranes may enhance regenerative outcomes around rough implants, but clinicians should be cautious, as radiographic bone gain does not necessarily reflect true re-osseointegration, and complete defect resolution is rarely achievable.

Nevertheless, these clinical implications must be interpreted with caution. The available evidence is heterogeneous, often based on small studies with differing peri-implantitis definitions, inconsistent outcome measures, and a serious overall risk of bias. The evidence was rated as very low to low for all main outcomes. This means that while current data can guide clinical choices, they cannot provide definitive recommendations.

## Data availability statement

The original contributions presented in the study are included in the article/[Supplementary Material](#), further inquiries can be directed to the corresponding author.

## Author contributions

PG: Conceptualization, Formal analysis, Methodology, Data curation, Writing – original draft, Writing – review & editing, Investigation. CG: Formal analysis, Methodology, Data curation, Writing – review & editing. AS: Conceptualization, Formal analysis, Methodology, Data curation, Investigation, Writing – review & editing, Project administration. AZ: Supervision, Conceptualization, Methodology, Formal analysis, Data curation, Investigation, Writing – original draft, Writing – review & editing.

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## Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fdmed.2025.1661369/full#supplementary-material>

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## Appendix

### Appendix 1 Excluded studies.

| Authors and date                 | Study title                                                                                                                                                               | Reason for exclusion                                                                                                       |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Afrashtehfar et al. 2024 (45)    | Guided bone regeneration improves defect fill and reconstructive outcomes in 3-wall peri-implantitis defects                                                              | Implant type not taken into consideration                                                                                  |
| Astolfi et al. 2021 (46)         | Influence of removing or leaving the prosthesis after regenerative surgery in peri-implant defects: retrospective study: 32 clinical cases with 2–8 years of follow-up    | Outcomes reported not correlated to implant type                                                                           |
| Behneke et al. 2000 (47)         | Treatment of peri-implantitis defects with autogenous bone grafts: six-month to 3-year results of a prospective study in 17 patients                                      | Implant type not taken into consideration                                                                                  |
| Berglundh 2018 (48)              | Long- term outcome of surgical treatment of periimplantitis. A 2–11-year retrospective study                                                                              | Minimum study duration less than 3 years                                                                                   |
| Bianchini et al. 2020 (49)       | Implantoplasty enhancing peri-implant bone stability over a 3-year follow-up: a case series                                                                               | Implant type not taken into consideration                                                                                  |
| Bianchini et al. (2024) (50)     | Clinical and radiographic outcomes of resective surgery with adjunctive implantoplasty over a 6- to 11-year follow-up: a case series                                      | Implants treated with implantoplasty, which creates a modified surface texture differing from the original implant surface |
| Carcuac et al. 2016 (51)         | Adjunctive systemic and local antimicrobial therapy in the surgical treatment of peri-implantitis: a randomized controlled clinical trial                                 | Study duration less than 3 years                                                                                           |
| Chiang et al. 2024 (52)          | Operating microscope-assisted reconstructive strategy for peri-implantitis: A case series report                                                                          | Implant type not taken into consideration                                                                                  |
| Cortellini et al. 2021 (53)      | Papilla preservation and minimally invasive surgery for the treatment of peri-implant osseous defects. Clinical and radiographic outcomes of a 5-year retrospective study | Implant type not taken into consideration                                                                                  |
| Froum et al. 2012 (54)           | Successful management of peri-implantitis with a regenerative approach: a consecutive series of 51 treated implants with 3- to 7.5-year follow-up                         | Outcomes mentioned not correlated to implant surfaces                                                                      |
| Froum et al. 2015 (55)           | A regenerative approach to the successful treatment of peri-implantitis: a consecutive series of 170 implants in 100 patients with 2- to 10-year follow-up                | Implant type not taken into consideration                                                                                  |
| Khayat et al. 2024 (56)          | Bone regeneration following implantoplasty: a retrospective cohort study with long-term radiographic assessment                                                           | Implant type not taken into consideration                                                                                  |
| Lombardo et al. 2019 (57)        | Successful management of peri-implantitis around short and ultrashort single-crown implants: a case series with a 3-year follow-up                                        | Implant type not taken into consideration                                                                                  |
| Monje et al. 2022 (58)           | Principles of combined surgical therapy for the management of peri-implantitis                                                                                            | Incorrect study design                                                                                                     |
| Parma-Benfenati et al. 2020 (59) | Long-term outcome of surgical regenerative treatment of peri-implantitis: a 2- to 21-year retrospective evaluation                                                        | Study duration less than 3 years (varied for 2–21 years)                                                                   |
| Renvert et al. 2012 (60)         | Surgical therapy for the control of peri-implantitis                                                                                                                      | Incorrect study design                                                                                                     |
| Renvert et al. 2024 (61)         | The efficacy of reconstructive therapy in the surgical management of peri-implantitis: A 3-year follow-up of a randomized clinical trial                                  | Implant type not taken into consideration                                                                                  |
| Sarmiento et al. 2018 (62)       | Surgical alternatives for treating peri-implantitis                                                                                                                       | Implant type not taken into consideration                                                                                  |
| Schwarz et al. 2015 (63)         | Reentry after combined surgical resective and regenerative therapy of advanced peri-implantitis: a retrospective analysis of five cases                                   | Implant type not taken into consideration                                                                                  |
| Schwarz et al. 2014 (64)         | Combined surgical therapy of advanced peri-implantitis lesions with concomitant soft tissue volume augmentation. A case series                                            | Study duration less than 3 years                                                                                           |
| Schwarz et al. 2013 (65)         | Four-year follow-up of combined surgical therapy of advanced peri-implantitis evaluating two methods of surface decontamination                                           | Implants treated with implantoplasty, which creates a modified surface texture differing from the original implant surface |
| Schwarz et al. 2017 (66)         | Combined surgical therapy of advanced peri-implantitis evaluating two methods of surface decontamination: a 7-year follow-up observation                                  | Implants treated with implantoplasty, which creates a modified surface texture differing from the original implant surface |
| Wang et al. 2021 (67)            | Laser-assisted regenerative surgical therapy for peri-implantitis: A randomized controlled clinical trial                                                                 | Study duration less than 3 years                                                                                           |

## Appendix 2 Full search strategies.

| Database                    | Search Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PubMed (MEDLINE via PubMed) | <p>(Periimplantitis OR peri-implantitis OR peri implantitis OR periimplant OR peri-implant OR peri implant)</p> <p>AND</p> <p>(treatment outcome OR therapy OR surgical treatment OR regenerative OR regeneration OR tissue regeneration OR reconstructive surgery OR bone graft OR bone substitute OR membranes OR surgical flap OR open flap debridement OR resective OR implantoplasty OR surface decontamination)</p> <p>AND</p> <p>(surface characteristics OR surface roughness OR material characteristics OR titanium surface OR implant types OR implant surfaces OR surface topography OR surface analysis)</p> <p>AND</p> <p>(implant survival OR bone loss OR recurrence OR retreatment OR radiographic stability OR long-term OR 3 years OR follow-up)</p>                                                                                                                                         |
| Embase                      | <p>('periimplantitis'/exp OR periimplantitis OR 'peri-implantitis' OR 'peri implantitis' OR periimplant OR 'peri-implant' OR 'peri implant')</p> <p>AND</p> <p>('treatment outcome'/exp OR therapy OR 'surgical treatment'/exp OR 'regenerative therapy'/exp OR regeneration OR 'tissue regeneration' OR 'reconstructive surgery'/exp OR 'bone graft'/exp OR 'bone substitute'/exp OR membranes OR 'surgical flap' OR 'open flap debridement' OR resective OR implantoplasty OR 'surface decontamination')</p> <p>AND</p> <p>('surface property'/exp OR 'surface roughness'/exp OR 'material property'/exp OR 'titanium surface' OR 'implant type'/exp OR 'implant surface'/exp OR 'surface topography'/exp OR 'surface analysis')</p> <p>AND</p> <p>('dental implant survival'/exp OR 'bone loss'/exp OR recurrence OR retreatment OR 'radiographic stability' OR 'long term' OR '3 years' OR 'follow-up')</p> |
| Cochrane library            | <p>(periimplantitis OR "peri-implantitis" OR "peri implantitis" OR periimplant OR "peri-implant" OR "peri implant")</p> <p>AND</p> <p>("treatment outcome" OR therapy OR "surgical treatment" OR regenerative OR regeneration OR "tissue regeneration" OR "reconstructive surgery" OR "bone graft" OR "bone substitute" OR membranes OR "surgical flap" OR "open flap debridement" OR resective OR implantoplasty OR "surface decontamination")</p> <p>AND</p> <p>("surface characteristics" OR "surface roughness" OR "material characteristics" OR "titanium surface" OR "implant types" OR "implant surfaces" OR "surface topography" OR "surface analysis")</p> <p>AND</p> <p>("implant survival" OR "bone loss" OR recurrence OR retreatment OR "radiographic stability" OR "long term" OR "3 years" OR "follow-up")</p>                                                                                   |